

Cardiovascular Health and Measurement Engineering

Physiology, Pathophysiology, Determinants, Evidence, Devices, Validation, and Buildable
Noninvasive Systems

Technical Evidence and Engineering Review

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Executive Summary

Cardiovascular health is not a single organ score. It is the emergent behavior of an electrical pump, compliant valves, elastic and muscular vessels, blood, kidneys, lungs, endocrine signals, autonomic control, immune biology, metabolism, behavior, and the surrounding environment. A useful cardiovascular profile must therefore combine **structure, electrical rhythm, pressure, flow, mechanical function, oxygen delivery, vascular state, biochemistry, and longitudinal context**. No one sensor - and no one laboratory value - can fully characterize it.

Cardiovascular disease remains the leading global cause of death. Depending on dataset, case definition, and year, recent estimates place annual cardiovascular deaths near 19 to 20 million worldwide. The World Health Organization estimated 19.8 million deaths in 2022, about 32% of all deaths, while global burden analyses report a continuing rise in total cardiovascular disability and deaths as populations grow and age (1–3). The most important interpretation is not the exact last decimal place: cardiovascular disease is common, begins biologically long before symptoms, and is substantially preventable or delayable.

The modern causal picture is unusually strong for several exposures. Cumulative exposure to atherogenic apolipoprotein B-containing particles, especially LDL particles; elevated blood pressure; tobacco smoke; hyperglycemia and diabetes; kidney disease; central adiposity and adverse metabolic state; and low physical capacity all have convergent evidence from physiology, prospective cohorts, genetics, randomized trials, and treatment-response studies. Their effects interact across decades rather than operating as isolated switches (4–6).

the phrase “**genetics versus environment**” **does not have one universal percentage answer**. Heritability asks how much variation within a particular population and environment is statistically associated with inherited differences; it does not ask what fraction of an individual’s disease was “caused by genes.” Coronary artery disease heritability is often estimated in the broad range of roughly 40% to 60% in family and twin studies, but estimates vary by ancestry, age, phenotype, and environment. Rare variants can be highly penetrant, as in familial hypercholesterolemia or some cardiomyopathies, whereas polygenic scores summarize thousands or millions of small effects. Yet high genetic risk and modifiable exposure are not mutually exclusive: healthy behavior is associated with substantially lower coronary risk even in people with high polygenic risk (7–9).

Population-attributable estimates answer a different question. The INTERHEART case-control study suggested that nine modifiable factors accounted for more than 90% of the population-attributable risk of a first myocardial infarction across studied populations, while a newer global consortium estimated that five major factors - body mass index, systolic pressure, non-HDL cholesterol, smoking, and diabetes - explained roughly half of incident cardiovascular disease, with the exact proportion differing by sex and region (6,10). These figures are not contradictory. They examine different endpoints, models, populations, and assumptions. They also do not imply that eliminating one factor would simply subtract its reported percentage because exposures overlap and interact.

The strongest practical conclusion is therefore a **life-course, cumulative-exposure model**:

- genes influence baseline susceptibility, lipid biology, pressure regulation, thrombosis, myocardial structure, and response to exposures;
- environments determine the distribution of food, activity, sleep, pollution, stress, infection, social resources, and care;
- behavior partly mediates environment but is constrained by biology and social conditions;
- metabolic and renal systems amplify or buffer those exposures;
- subclinical damage accumulates silently until reserve is lost or an acute trigger exposes the underlying substrate.

Current prevention guidance reflects this integrated view. The 2025 US blood-pressure guideline uses PREVENT risk estimates, home measurement, lifestyle therapy, and a general treatment goal below 130/80 mmHg for most adults while individualizing care. The 2026 dyslipidemia guideline restores explicit LDL-C and non-HDL-C goals by risk category, recommends wider use of apolipoprotein B and at least once-in-lifetime lipoprotein(a) measurement, and emphasizes earlier lifetime-risk reduction. The first 2026 cardiovascular-kidney-metabolic guideline treats obesity, diabetes, kidney disease, and cardiovascular disease as an interconnected syndrome rather than separate silos. The 2026 AHA dietary statement emphasizes an overall pattern rich in vegetables, fruits, whole grains, legumes, nuts, fish, and unsaturated fats while minimizing sodium, added sugars, and ultraprocessed foods; it does not recommend starting alcohol for health (11–14).

For device engineering, the central rule is **measure the measurand, not the marketing label**. An ECG measures voltage differences at the body surface, not cardiac output. Photoplethysmography measures optical changes related

to local pulsatile blood volume, not blood pressure. An oscillometric cuff senses pressure oscillations in a compressed artery and applies an algorithm to estimate systolic and diastolic pressure; mean arterial pressure is usually closer to the direct envelope maximum, while systolic and diastolic values are model-dependent. Pulse-transit time is influenced by arterial stiffness and pre-ejection period and cannot, by itself, provide universal calibration-free blood pressure. Echocardiographic ejection fraction is a volumetric ratio with loading dependence, not a complete measure of contractility. Troponin indicates myocardial injury, not necessarily infarction. Every device has a **measurement chain**, an **inference layer**, and a **clinical interpretation layer**; engineering claims must identify which layer has actually been validated.

This packet develops a complete measurement landscape:

1. **Electrical:** resting 12-lead ECG, ambulatory patches, Holter monitors, event recorders, implantable loop recorders, telemetry, body-surface mapping, intracardiac electrograms, pacemaker/ICD sensing, and electrophysiology systems.
2. **Pressure and hemodynamics:** manual and automated cuffs, ambulatory blood pressure, finger volume-clamp systems, tonometry, pulse-wave analysis, invasive arterial and venous pressure, pulmonary artery catheters, left-heart catheterization, pressure-volume loops, fractional flow reserve, instantaneous wave-free ratio, and implantable pulmonary artery pressure monitoring.
3. **Optical:** reflective and transmissive photoplethysmography, pulse oximetry, remote camera PPG, near-infrared spectroscopy, laser Doppler, laser speckle, and retinal vascular imaging.
4. **Mechanical and acoustic:** phonocardiography, electronic stethoscopes, seismocardiography, ballistocardiography, gyrocardiography, impedance cardiography, and wearable ultrasonics.
5. **Imaging and flow:** echocardiography, Doppler, strain, cardiac CT, coronary calcium, CT angiography, CMR, SPECT, PET, catheter angiography, intravascular ultrasound, optical coherence tomography, vascular ultrasound, ankle-brachial index, pulse-wave velocity, and endothelial function testing.
6. **Biochemical and functional:** lipids, apoB, Lp(a), glucose, HbA1c, kidney markers, troponin, natriuretic peptides, inflammation markers, exercise ECG, stress imaging, six-minute walk, and cardiopulmonary exercise testing.

The buildable systems proposed here are intentionally noninvasive and staged. The recommended first platform is a synchronized, battery-powered research unit combining one- or two-channel ECG, dual-wavelength PPG, a low-noise inertial sensor, contact temperature, and optional acoustic or bioimpedance channels. It should log raw data, preserve exact timestamps, expose quality indices, and be validated against reference instruments before any health interpretation. A second platform adds a validated upper-arm cuff or a research cuff system with a pressure transducer and controlled pneumatic path. A third benchtop station adds multi-lead ECG, digital phonocardiography, bilateral limb PPG or Doppler for vascular timing, and standardized exercise/recovery protocols. These systems can capture rhythm, rate variability, pulse morphology, electromechanical timing, motion, posture, temperature, respiratory modulation, and trends. They cannot replace echocardiography, coronary imaging, laboratory testing, or invasive hemodynamics.

This packet is an educational and engineering reference, not personal medical advice and not authorization to use a prototype for diagnosis or treatment. Human-connected prototypes should remain battery powered while attached, should never have a direct conductive path to mains-powered equipment or an unisolated USB host, and should be tested first with physiological simulators and protection networks. Any invasive, implantable, defibrillation-connected, catheter-based, or treatment-delivering system requires licensed clinical teams, formal risk management, ethics oversight, and applicable regulatory controls. Do not build or use an invasive device from this document.

What this packet is designed to answer

- What does a healthy cardiovascular system do mechanically and electrically?
- How do atherosclerosis, hypertension, ischemia, heart failure, arrhythmias, valve disease, vascular disease, and thrombosis arise?
- Which risk factors are causal, which are useful markers, and which remain uncertain?
- How should genetics, nutrition, exercise, stress, sleep, tobacco, alcohol, pollution, and social conditions be weighted without false precision?

Cardiovascular health is a coupled system

Electrical activation, mechanical pumping, vascular transport, and metabolic control are inseparable

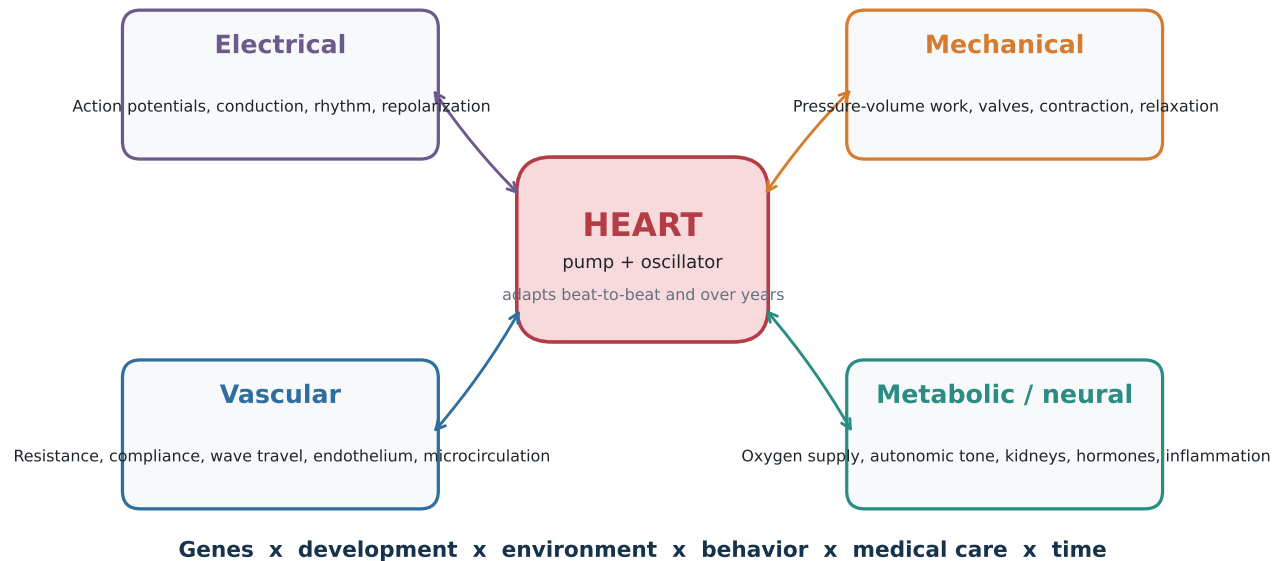


Figure 1: Cardiovascular health is a coupled system of pump, vessels, blood, control, metabolism, kidneys, lungs, behavior, and environment.

- What does each clinical or wearable device actually measure?
- Which noninvasive signals can be combined to build a credible longitudinal profile?
- What are the analog, digital, algorithmic, safety, regulatory, and validation requirements for building such devices?
- Where are the limits of current science, especially for cuffless blood pressure, consumer arrhythmia detection, AI interpretation, and biomarker-free claims?

How to Use This Packet

The document is organized from first principles to implementation.

Parts I and II explain the normal system and major disease processes. **Part III** evaluates determinants and the relative contributions of inherited, behavioral, metabolic, environmental, and social factors. **Part IV** summarizes prevention evidence and current guidance. **Part V** is a modality-by-modality measurement atlas. **Part VI** turns that landscape into engineering requirements, reference architectures, signal-processing pipelines, and validation programs. **Part VII** defines integrated assessment tiers and a practical research roadmap. Appendices collect equations, signal specifications, reference intervals, standards, protocol checklists, and a glossary.

Evidence is described using four informal levels:

Label	Meaning	Typical evidence
Established causal	Multiple independent methods support a causal contribution, and modifying the factor changes outcomes or validated intermediates.	Randomized trials, Mendelian randomization, mechanistic data, dose-response, consistency.
Probable/important	Strong and coherent evidence, but intervention data or measurement precision may be incomplete.	Prospective cohorts, natural experiments, mechanistic studies, partial trial evidence.
Risk marker	Predicts disease but may not be causal or modifiable.	Reproducible association and discrimination; causal pathway uncertain.
Unsettled	Results are inconsistent, confounded, endpoint-dependent, or insufficiently validated.	Small studies, post hoc analyses, surrogate-only results, nonrepresentative datasets.

The same discipline is applied to devices:

Layer	Question
Physical measurand Transduction	What physical or chemical quantity reaches the sensor? How is it converted to voltage, current, digital code, image, or waveform?
Estimation	Which hidden physiological variable is inferred, and from what model or calibration?
Clinical claim Validation	What diagnosis, prognosis, or treatment decision is claimed? Against which reference, population, protocol, and error criterion was the claim tested?

Part I

Foundations: History, Physiology, and Measurement Logic

Historical Development of Cardiovascular Science

The cardiovascular field advanced whenever a new instrument made an invisible variable observable. The history is therefore not just a sequence of discoveries; it is a sequence of **measurement revolutions**.

1.1 From pulse observation to a closed circulation

Ancient physicians described the pulse and related its qualities to illness, but the dominant models of blood movement were not a closed-loop circulation. William Harvey's seventeenth-century experiments combined anatomy, ligature tests, valve observations, and quantitative reasoning to show that the heart continuously propels blood through a circulation. His argument was partly an engineering mass-balance argument: the liver could not plausibly create the enormous quantity of blood that would be required if blood were continually consumed rather than recirculated.

Stephen Hales made the first direct arterial pressure measurements in animals in the early eighteenth century using a vertical glass tube connected to an artery. That experiment made pressure a measurable state variable. Poiseuille later quantified laminar flow through tubes, and Hagen-Poiseuille behavior became a useful - though simplified - model of resistance in small rigid vessels. The later development of fluid-filled manometers, elastic theory, and wave analysis made clear that arteries are not rigid pipes and blood flow is pulsatile, branching, and non-Newtonian at some scales.

1.2 Auscultation, cuff pressure, and the clinical examination

Laennec's stethoscope turned internal acoustics into a bedside signal. Riva-Rocci's inflatable cuff and Korotkoff's auscultatory sounds transformed arterial pressure into a reproducible noninvasive measurement. These devices illustrate a theme that remains essential: the measured signal is indirect. Korotkoff sounds emerge from flow transitions and arterial wall behavior under external compression; the cuff does not directly sense central aortic pressure.

The physical examination itself became a multisensor fusion system: palpation of pulses, observation of jugular venous motion, auscultation of heart sounds and bruits, inspection for edema or cyanosis, and measurement of pressure and respiratory response. Modern imaging and electronics refine rather than eliminate these physiologic observations.

1.3 Electricity and the electrocardiogram

The discovery that excitable tissue produces electrical currents led from capillary electrometers to Waller's early human recordings and Willem Einthoven's string galvanometer. Einthoven formalized limb leads and the triangle relation; Wilson, Goldberger, and others expanded the system to unipolar and augmented leads, producing the modern 12-lead ECG. The ECG became a powerful example of field measurement: electrodes do not record the membrane action potential of one myocyte. They record time-varying potential differences on the body surface generated by the integrated vector of depolarizing and repolarizing myocardium filtered through the volume conductor.

Continuous monitoring followed with Holter recording, bedside telemetry, implantable pacemakers, defibrillators, event recorders, and loop recorders. Digital signal processing enabled automated interval measurement, rhythm classification, compression, remote transmission, and now machine-learning inference. Each step expanded duration and accessibility but also increased the importance of artifact detection and false-positive management (15–17).

1.4 Catheterization and direct hemodynamics

Werner Forssmann's self-catheterization, followed by the work of Cournand and Richards, established cardiac catheterization as a clinical science. Pressure could be sampled in chambers and vessels, oxygen saturations could reveal shunts, cardiac output could be estimated by Fick or indicator-dilution methods, and angiography could define anatomy. Selective coronary angiography, right-heart catheterization, pressure wires, intravascular ultrasound, and optical coherence tomography progressively connected anatomy to physiology.

The Swan-Ganz pulmonary artery catheter provided continuous right-sided and pulmonary pressures and thermodilution cardiac output. It also demonstrated an important lesson: **more measurement does not automatically improve outcomes**. In broad heart-failure populations, routine catheter use did not improve outcomes when not tied to an effective decision pathway, emphasizing that a monitor is useful only when accurate data lead to effective action (18).

1.5 Imaging revolutions

M-mode and two-dimensional echocardiography made moving structure visible without ionizing radiation. Doppler added velocity and pressure-gradient estimation; tissue Doppler and speckle tracking added myocardial motion and deformation. Nuclear perfusion imaging linked tracer distribution to flow and metabolism. CT developed from anatomic cross-sections to rapid coronary calcium scoring and coronary CT angiography. Cardiovascular magnetic resonance combined anatomy, function, flow, edema, perfusion, and tissue characterization. Intravascular ultrasound and optical coherence tomography moved imaging inside coronary arteries.

No modality superseded all others because each samples a different combination of spatial resolution, temporal resolution, penetration, tissue contrast, physiology, risk, cost, and portability. The engineering objective is not to identify a universally best sensor, but to match the modality to the question.

1.6 Population science and prevention

The Framingham Heart Study transformed cardiovascular medicine by identifying and quantifying risk factors longitudinally. Cohorts such as the Seven Countries Study, Nurses' Health Study, Whitehall studies, ARIC, MESA, UK Biobank, and many others expanded the biological and social landscape. Randomized trials then established that lowering blood pressure, lowering LDL cholesterol, stopping smoking, using selected metabolic therapies, and treating specific disease states reduces events.

Modern prevention has evolved from treating isolated thresholds to estimating total and lifetime risk, identifying subclinical disease, and treating cardiovascular-kidney-metabolic interactions (13,19). Wearables now create population-scale longitudinal datasets, but those data introduce new biases: device ownership, adherence, skin contact, motion, firmware changes, missingness, and selection into follow-up.

1.7 A concise measurement timeline

Era	Enabling measurement	New physiological question
1600s	Quantitative anatomy and ligature experiments	Does blood circulate in a closed loop?
1700s	Direct fluid-column manometry	What pressure does the heart generate?
1800s	Stethoscope, sphygmograph, early electrical recording	What sounds, pulse shapes, and fields accompany the cardiac cycle?
Early 1900s	Cuff sphygmomanometry and clinical ECG	Can pressure and electrical abnormalities be measured noninvasively?
Mid 1900s	Catheterization, angiography, Holter monitoring	What are chamber pressures, flows, anatomy, and intermittent rhythms?
1950s-1980s	Echo, nuclear imaging, bypass, pacemakers	Can structure, function, perfusion, and rhythm be treated and monitored?

Era	Enabling measurement	New physiological question
1990s-2010s	Digital ECG, CT, CMR, IVUS/OCT, implantable monitors	Can anatomy, tissue, and continuous physiology be integrated?
2010s-2026	Wearables, cloud telemetry, multimodal AI	Can longitudinal population monitoring detect risk earlier without creating unacceptable false alarms?

Cardiovascular System Architecture

The cardiovascular system can be represented as two pumps in series - right and left heart - connected to two vascular beds - pulmonary and systemic - with parallel organ branches within the systemic circulation. This simple topology is complicated by time-varying elastance, valves, compliant vessels, autoregulation, blood rheology, respiration, neural and hormonal control, and changing metabolic demand.

2.1 Anatomical components

The right atrium receives systemic venous return and transfers blood to the right ventricle. The right ventricle pumps into the low-pressure pulmonary circulation. Oxygenated blood returns to the left atrium and enters the left ventricle, which pumps into the high-pressure systemic circulation. Four valves impose directional flow: tricuspid, pulmonic, mitral, and aortic. Coronary arteries arise from the aortic root and perfuse myocardium; most left-ventricular coronary flow occurs during diastole because systolic contraction compresses intramyocardial vessels.

Arteries buffer pulsatile ejection, arterioles dominate adjustable resistance, capillaries exchange gases and solutes, venules and veins serve as a large capacitance reservoir, and lymphatic vessels return interstitial fluid and macromolecules. The kidneys regulate volume, sodium, pressure, and endocrine signals; lungs set gas exchange and influence right-heart load; the liver and adipose tissue shape lipid and metabolic state; bone marrow and immune organs influence inflammation and thrombosis.

2.2 Conservation relationships

At steady state, average right- and left-ventricular outputs must be equal over time. Short-lived differences are absorbed by changes in pulmonary or systemic blood volume, but persistent imbalance produces congestion or underfilling.

Cardiac output is

$$CO = HR \times SV$$

where HR is heart rate and SV is stroke volume. Stroke volume is

$$SV = EDV - ESV$$

and ejection fraction is

$$EF = \frac{SV}{EDV} \times 100\%.$$

Ejection fraction is useful but incomplete. It depends on preload and afterload; a small, thick-walled ventricle can have a normal or high EF despite impaired filling and low stroke volume, while a dilated ventricle can have a low EF yet maintain output through a large end-diastolic volume.

A hydraulic analogue of Ohm's law is

$$\Delta P = QR,$$

where pressure difference drives mean flow Q through resistance R . For the systemic circulation,

$$MAP - RAP \approx CO \times SVR,$$

where MAP is mean arterial pressure, RAP is right-atrial pressure, and SVR is systemic vascular resistance. This is a mean-flow approximation; instantaneous arterial behavior also depends on compliance, inertance, and wave propagation.

2.3 Oxygen delivery and extraction

Systemic oxygen delivery is

$$DO_2 = CO \times C_{aO_2},$$

where arterial oxygen content is approximately

$$C_{aO_2} = 1.34 \times Hb \times S_{aO_2} + 0.003 \times P_{aO_2}.$$

The hemoglobin-bound term dominates under ordinary conditions. Oxygen consumption follows the Fick principle:

$$\dot{V}O_2 = CO \times (C_{aO_2} - C_{vO_2}).$$

This equation underlies direct Fick cardiac output during catheterization and explains why heart rate or oxygen saturation alone cannot quantify systemic performance. A person may maintain resting pressure and saturation while using more extraction, higher sympathetic tone, or lower reserve.

2.4 Control loops

Fast regulation operates through arterial and cardiopulmonary baroreflexes, autonomic output, and local vascular responses. Intermediate regulation includes the renin-angiotensin-aldosterone system, vasopressin, natriuretic peptides, endothelial mediators, and renal pressure-natriuresis. Long-term regulation depends heavily on renal sodium handling, vascular remodeling, metabolic state, and structural cardiac adaptation.

A single reading is therefore a sample from a dynamic controlled system. Pressure varies with posture, respiration, temperature, emotion, recent activity, cuff technique, medication timing, and circadian phase. Heart-rate variability varies with breathing, sleep, age, fitness, illness, and signal-processing choices. Good monitoring treats context as data rather than noise.

Electrical Physiology and the Surface ECG

3.1 Cellular action potentials

Cardiac electrical behavior arises from ion gradients and voltage-gated channels. Pacemaker cells in the sinoatrial node depolarize spontaneously through a balance of funny current, calcium currents, potassium currents, and intracellular calcium cycling. Working atrial and ventricular myocytes have a rapid sodium-mediated upstroke, early repolarization, a calcium-supported plateau, potassium-mediated repolarization, and a stable resting potential. Nodal tissue has slower calcium-dependent conduction and a longer decremental delay.

The effective refractory period prevents immediate re-excitation and normally blocks sustained tetany. Reentry requires a circuit, unidirectional block, and conduction time long enough for tissue ahead of the wavefront to recover. Triggered activity can arise from early or delayed afterdepolarizations. Automaticity, triggered activity, and reentry form a useful framework for arrhythmia mechanisms.

3.2 Conduction sequence

The normal impulse begins in the sinoatrial node, spreads through the atria, pauses in the atrioventricular node, traverses the His bundle, bundle branches, and Purkinje network, then depolarizes ventricular myocardium from endocardium toward epicardium in a coordinated sequence. Repolarization follows a different spatial sequence, so the T wave is not simply an inverted QRS.

Electrical activation creates a body-surface voltage projection - not a direct image of contraction

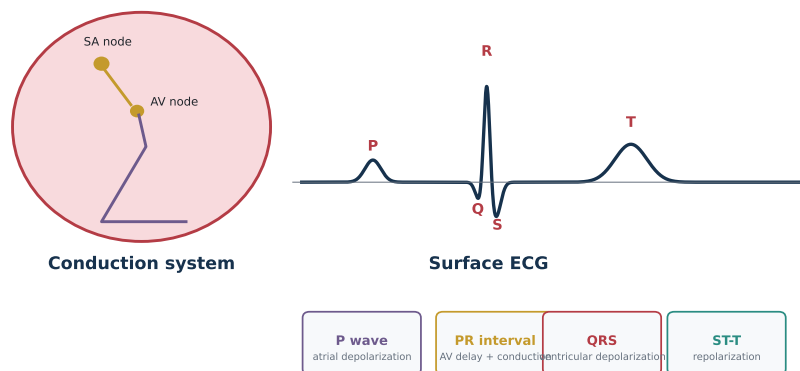


Figure 3.1: Relationship between conduction, mechanical events, and a schematic ECG. Surface ECG timing is related to, but not identical with, mechanical timing.

3.3 What the ECG measures

An ECG lead is a voltage difference between electrodes or a derived virtual reference. A 12-lead ECG uses ten physical electrodes to produce twelve standardized views: three bipolar limb leads, three augmented limb leads, and six precordial leads. The lead system projects a three-dimensional, time-varying cardiac source onto defined axes.

Typical diagnostic content includes:

- **Rate and rhythm:** sinus rhythm, atrial fibrillation, flutter, tachycardia, bradycardia, ectopy.
- **Conduction:** PR interval, QRS duration, atrioventricular block, bundle-branch block, pre-excitation.
- **Repolarization:** ST deviation, T-wave change, QT/QTc, electrolyte or drug effects.
- **Chamber and axis clues:** hypertrophy patterns, axis deviation, prior infarct patterns, though sensitivity and specificity vary.
- **Temporal behavior:** pauses, runs, burden, rate response, symptom-rhythm correlation.

The ECG cannot directly measure coronary stenosis, ejection fraction, cardiac output, filling pressure, or plaque burden. It can be normal despite serious structural or coronary disease, and abnormal without immediate danger.

3.4 Intervals and corrections

Common intervals are defined from fiducial points:

- PR: atrial depolarization onset to ventricular depolarization onset.
- QRS: duration of ventricular depolarization.
- QT: ventricular depolarization through repolarization.
- RR: interval between successive ventricular activations.

QT varies with rate. Bazett's formula, $QT_c = QT/\sqrt{RR}$, is widely used but overcorrects at high rates and undercorrects at low rates. Fridericia, Framingham, Hodges, and individualized corrections can perform better in different contexts. Device software should preserve the uncorrected QT, RR, correction formula, confidence, and representative beats rather than outputting only one opaque QTc number.

3.5 Signal bandwidth and morphology

Diagnostic ECG requires preservation of low-frequency ST content and high-frequency QRS detail. Monitoring systems may use narrower bandwidths to suppress baseline wander and muscle noise, but this can distort morphology. A research device should distinguish a **diagnostic morphology mode** from a **rhythm-monitoring mode** and disclose filters. Zero-phase offline filters can preserve waveform timing but are noncausal; real-time filters create delay and phase response. Filtering should never be used to make a poor electrode interface appear clinically clean.

3.6 Electromechanical separation

The electrical impulse initiates contraction but does not equal contraction. The interval from QRS onset to aortic valve opening includes electrical activation and electromechanical delay. Pulse arrival at the wrist or finger adds vascular transit time. This distinction is central to pulse-transit approaches: ECG-to-PPG delay includes pre-ejection period plus arterial transit and is not pure pulse transit.

Measurement principle

Electrical, mechanical, pressure, acoustic, and optical signals are different projections of the same cardiac cycle. Their timing relationships are often more informative than any single waveform, but only when channels are synchronized to a common clock and sensor-specific delays are characterized.

Mechanical Physiology and the Cardiac Cycle

4.1 Pressure-volume analysis

The left-ventricular pressure-volume loop is a compact representation of one cardiac cycle. It separates filling, isovolumic contraction, ejection, and isovolumic relaxation. Loop width is stroke volume; loop area approximates external stroke work. End-systolic pressure-volume relationship is often used as a load-insensitive index of contractile state over a limited range, while the end-diastolic pressure-volume relationship reflects passive chamber stiffness.

Pressure-volume loop: one compact view of ventricular mechanics

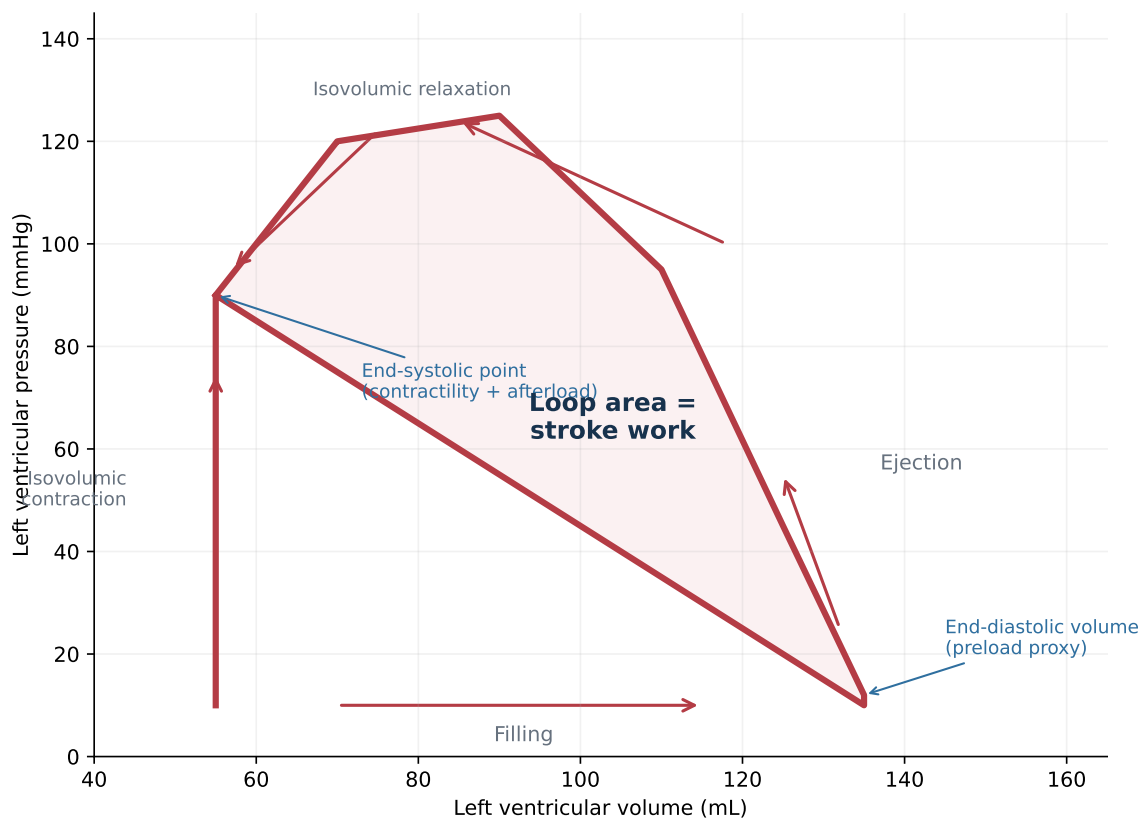


Figure 4.1: Schematic pressure-volume loop showing phases, valve events, stroke volume, preload, afterload, and contractility-related shifts.

Preload is the initial myocardial stretch or filling condition, often approximated imperfectly by end-diastolic volume or pressure. **Afterload** is the load opposing ejection; it is not synonymous with brachial systolic pressure and includes arterial pressure, impedance, wave reflection, ventricular geometry, and wall stress. **Contractility** is intrinsic active force-generating state independent of loading, though no simple noninvasive measure is perfectly load-independent. **Lusitropy** describes relaxation.

4.2 Frank-Starling mechanism

Within physiological limits, increased end-diastolic fiber length increases contraction force and stroke volume. This allows the heart to match venous return without a central controller commanding each beat. The relation shifts

with contractility and is constrained by pericardial interaction, ventricular compliance, valve disease, and right-left coupling. In heart failure, higher filling pressures may produce little additional output and substantial congestion.

4.3 Valve mechanics and heart sounds

The first heart sound is associated mainly with closure and tensioning of the mitral and tricuspid apparatus at the onset of systole. The second is associated with aortic and pulmonic valve closure. Additional sounds can arise from rapid filling, atrial contraction into a stiff ventricle, prosthetic valves, or abnormal flow. Murmurs reflect turbulent or disturbed flow and are characterized by timing, location, radiation, frequency, intensity, and response to maneuvers.

Acoustic energy depends on flow, pressure gradient, orifice geometry, vessel or chamber resonance, tissue transmission, sensor contact, and ambient noise. A digital stethoscope therefore measures a body-surface acoustic signal, not valve area. Valve severity is established by integrated imaging and hemodynamic criteria.

4.4 Right-heart physiology

The right ventricle is thin-walled and adapted to a low-resistance pulmonary circuit. It is highly sensitive to acute afterload increases. Pulmonary hypertension, pulmonary embolism, hypoxic vasoconstriction, mechanical ventilation, or left-heart disease can raise right-ventricular load. Right-ventricular dilation shifts the interventricular septum and can reduce left-ventricular filling through ventricular interdependence.

Respiration strongly modulates right-sided filling and venous pressure. Intrathoracic pressure changes affect transmural pressure, venous return, and measured catheter values. Any device estimating filling pressure or volume from thoracic impedance, jugular waveforms, or pulse morphology must model posture and respiration explicitly.

Vascular Mechanics, Pressure, Flow, and Waves

5.1 Resistance and Poiseuille intuition

For steady laminar flow in a long rigid circular tube,

$$R = \frac{8\mu L}{\pi r^4}.$$

The fourth-power dependence on radius explains why arterioles can control resistance effectively. Real vessels are elastic, tapering, branching, actively regulated, and exposed to pulsatile flow; blood is a suspension whose effective viscosity changes with vessel size and shear. The equation is therefore an intuition, not a complete cardiovascular model.

5.2 Compliance and the Windkessel

Arterial compliance is

$$C = \frac{\Delta V}{\Delta P}.$$

Elastic arteries store part of stroke volume during systole and return it during diastole, smoothing flow. In a two-element Windkessel model, resistance and compliance determine diastolic pressure decay. More complete models add characteristic impedance and inertance. With age, hypertension, calcification, and structural remodeling, arteries stiffen, pulse-wave velocity rises, systolic pressure and pulse pressure often increase, and reflected waves may return earlier.

5.3 Pulse-wave propagation

A pressure wave travels far faster than the blood itself. In simplified form, the Moens-Korteweg relation gives wave speed

$$PWV \approx \sqrt{\frac{Eh}{\rho D}},$$

where E is effective elastic modulus, h wall thickness, D diameter, and ρ blood density. The Bramwell-Hill relation expresses wave speed in terms of distensibility. Both are idealizations, but they show why stiffness, geometry, and pressure state influence transit time.

Wave reflections arise at impedance changes such as branches, taper, and resistance vessels. Peripheral systolic pressure can exceed central systolic pressure because of amplification. Tonometry and pulse-wave analysis attempt to estimate central pressure or augmentation, but results depend on calibration, transfer functions, sensor placement, and population.

5.4 Mean pressure and pulse pressure

At ordinary heart rates, mean arterial pressure is often approximated as

$$MAP \approx DBP + \frac{1}{3}(SBP - DBP),$$

but the coefficient changes with heart rate and waveform shape. Direct numerical integration of a valid arterial waveform is preferable. Pulse pressure is $SBP - DBP$ and reflects the interaction of stroke volume, arterial compliance, ejection pattern, and wave reflection rather than one isolated property.

5.5 Endothelium and microcirculation

The endothelium regulates tone, permeability, inflammation, thrombosis, and remodeling through nitric oxide, prostacyclin, endothelin, reactive oxygen species, surface receptors, and paracrine signals. Shear stress influences endothelial phenotype. Microvascular function determines tissue perfusion and can be abnormal even when epicardial coronary arteries are unobstructed.

Clinical and research measures such as brachial flow-mediated dilation, peripheral arterial tonometry, coronary flow reserve, retinal vessel analysis, laser Doppler, and reactive hyperemia each interrogate different vascular beds and mechanisms. They should not be treated as interchangeable generic “endothelial function” scores (20,21).

Part II

Pathophysiology: How Cardiovascular Disease Develops

Atherosclerosis, Plaque Biology, and Thrombosis

Atherosclerosis is a chronic, lipid-driven, inflammatory disease of the arterial wall. It develops over decades and can remain clinically silent until flow limitation, plaque disruption, thrombosis, embolization, or loss of vascular reserve causes symptoms.

6.1 Initiation and retention of apoB particles

Apolipoprotein B-containing particles - LDL, remnant particles, and lipoprotein(a) - enter the arterial intima. Retention through interactions with proteoglycans is a key initiating event. Particles undergo modification, trigger endothelial activation, and recruit monocytes. Macrophages ingest lipids and become foam cells. Smooth-muscle cells migrate, proliferate, produce extracellular matrix, and can adopt macrophage-like phenotypes. Repeated injury and repair produce a heterogeneous lesion (4,5,22).

The causal exposure is better represented by the **number and duration of atherogenic particles** than by cholesterol concentration alone. LDL-C measures cholesterol mass carried in LDL particles; apoB approximates the number of atherogenic particles because each particle carries one apoB molecule. Discordance occurs when particles are cholesterol-poor or cholesterol-rich. Non-HDL-C captures cholesterol in all apoB particles. Lp(a) adds genetically determined apoB-containing particles with proatherogenic and prothrombotic features (12,23).

6.2 Plaque progression and phenotype

Plaques can enlarge outward before narrowing the lumen, so a normal stress test or angiographic lumen does not exclude substantial wall disease. Fibrous cap thickness, lipid/necrotic core, inflammation, neovascularization, intraplaque hemorrhage, calcification pattern, and local mechanics influence vulnerability. “Stable” and “vulnerable” are not permanent categories; biology evolves.

Acute coronary syndromes can result from fibrous-cap rupture, superficial erosion, or less often calcified nodules. Thrombosis depends on exposed tissue factor, platelets, coagulation, endogenous fibrinolysis, flow, and systemic inflammatory or sympathetic state. Severe fixed stenosis can cause demand ischemia without acute plaque disruption; nonobstructive plaques can still cause infarction if they thrombose.

6.3 Why LDL lowering works

Concordant evidence comes from familial hypercholesterolemia, common variants, Mendelian randomization, observational exposure, statins, ezetimibe, PCSK9 inhibitors, and other therapies. The reduction in major vascular events is approximately proportional to the absolute reduction in LDL-C and duration of treatment, with larger lifetime benefit when exposure is lowered earlier (24–30).

This does not imply that LDL is the only cause. Blood pressure, smoking, diabetes, inflammation, thrombosis, and vessel biology change the probability that a given plaque develops or causes an event. Atherosclerosis is multicausal, but apoB particle exposure is a necessary central driver for the common atherosclerotic process.

6.4 Inflammation: causal but not nonspecific

Inflammation participates in plaque initiation, progression, and complications. CANTOS showed that targeted interleukin-1 beta inhibition reduced recurrent cardiovascular events without lowering lipids, supporting a causal

inflammatory pathway, but infection risk and cost limited broad use. Low-dose colchicine trials showed event reduction in selected secondary-prevention populations, again supporting inflammatory modulation (31–33).

High-sensitivity C-reactive protein is a useful risk marker and reflects inflammatory state, but it is nonspecific. Lowering a marker is not automatically beneficial; the pathway and intervention matter.

6.5 Clinical manifestations

- coronary artery disease and myocardial infarction;
- ischemic stroke and transient ischemic attack;
- peripheral artery disease and limb ischemia;
- aortic and branch-vessel atherosclerosis;
- renovascular disease and mesenteric ischemia;
- microembolization and cholesterol emboli.

A complete profile therefore includes more than coronary symptoms: pulses, limb pressures, neurologic history, kidney function, vascular imaging when indicated, and systemic risk factors.

Hypertension and Pressure-Mediated Injury

Blood pressure is both a regulated physiological variable and a mechanical exposure delivered to vessels and organs with every beat. Risk rises continuously across the population; diagnostic categories are decision thresholds, not biological cliffs.

7.1 Origins of elevated pressure

Long-term arterial pressure emerges from cardiac output, vascular resistance, arterial stiffness, renal sodium handling, sympathetic tone, endocrine systems, body size, sleep, medications, and environment. Common primary hypertension is heterogeneous. Mechanisms include impaired pressure natriuresis, excessive sodium retention, renin-angiotensin-aldosterone activity, sympathetic activation, endothelial dysfunction, vascular remodeling, and stiffness.

Secondary causes include kidney parenchymal disease, renovascular disease, primary aldosteronism, obstructive sleep apnea, thyroid or adrenal disorders, coarctation, pregnancy-related disorders, and drugs or substances. Secondary evaluation is guided by age, severity, resistant pattern, abrupt onset, laboratory findings, and clinical context.

7.2 Organ damage

High pressure increases left-ventricular wall stress and promotes hypertrophy, fibrosis, diastolic dysfunction, atrial enlargement, and eventually heart failure or arrhythmia. It accelerates atherosclerosis, damages small arteries, contributes to ischemic and hemorrhagic stroke, promotes kidney disease, stiffens large arteries, and damages retina and cognition.

The relationship is bidirectional: kidney disease raises pressure, and pressure accelerates kidney disease. Obesity, sleep apnea, diabetes, and sympathetic activation create reinforcing loops. The 2026 CKM framework is explicitly designed around these interactions (13).

7.3 Measurement is part of the disease definition

Poor technique can create false hypertension or mask true hypertension. Cuff size, arm position, back and feet support, rest period, talking, recent caffeine or exercise, bladder distension, device validation, and repeated averaging matter. Office readings may differ from home and ambulatory pressure. White-coat hypertension and masked hypertension cannot be reliably identified from one office reading (11,34).

Ambulatory blood pressure adds daytime, nighttime, sleep dipping, morning surge, and variability. It is a stronger predictor than office pressure in many contexts. Home monitoring improves longitudinal assessment if technique and device validation are adequate.

7.4 Evidence from treatment trials

Randomized trials establish that lowering pressure reduces stroke, heart failure, and major cardiovascular events. SPRINT showed benefit from an intensive systolic target in selected high-risk adults without diabetes, using a specific standardized measurement protocol. ACCORD BP in diabetes did not meet its primary composite endpoint but reduced stroke and informed interpretation of population-specific benefit and adverse events. STEP supported intensive control in older adults in China. These trials should not be reduced to one universal number without considering measurement method, eligibility, frailty, orthostatic symptoms, adverse effects, and absolute risk (35–38).

Current US guidance generally targets below 130/80 mmHg in most adults and uses PREVENT risk plus comorbidity to decide when medication should accompany lifestyle therapy (11).

Ischemia, Myocardial Infarction, and Coronary Microvascular Disease

Myocardial ischemia occurs when oxygen supply is insufficient for demand. Supply depends on coronary flow, perfusion pressure, diastolic time, vascular resistance, oxygen content, and collateral circulation. Demand depends on heart rate, contractility, wall stress, and myocardial mass.

8.1 Epicardial obstruction and flow limitation

A fixed stenosis creates a pressure drop that grows with flow. Resting flow can remain normal because downstream arterioles dilate; flow reserve falls first. During exercise or pharmacologic hyperemia, demand exceeds maximal supply and ischemia develops. Fractional flow reserve estimates the ratio of maximal flow in a diseased artery to hypothetical maximal flow without the stenosis, using distal-to-aortic pressure during hyperemia. Instantaneous wave-free ratio uses a resting diastolic period. Both are invasive physiological indices and differ from visual percent stenosis (39,40).

8.2 Acute myocardial infarction

The universal definition classifies infarction by mechanism and clinical setting. Type 1 usually reflects acute atherothrombosis. Type 2 reflects oxygen supply-demand imbalance without acute plaque thrombosis, such as severe anemia, hypoxemia, tachyarrhythmia, shock, or hypertensive crisis. Other categories relate to sudden death, PCI, stent thrombosis, or bypass surgery.

Troponin elevation establishes myocardial injury when above the assay's 99th percentile and changing appropriately; infarction additionally requires evidence of ischemia. Myocarditis, heart failure, kidney disease, pulmonary embolism, sepsis, and strenuous exertion can elevate troponin. A sensor that detects elevated troponin would detect injury biology, not the mechanism.

8.3 Ischemia without obstructive coronary arteries

Angina or infarction can occur with nonobstructive epicardial arteries. Mechanisms include microvascular dysfunction, vasospasm, endothelial dysfunction, embolism, spontaneous coronary dissection, supply-demand mismatch, or recanalized thrombosis. Coronary flow reserve and microvascular resistance may be assessed invasively or by PET, CMR, or Doppler methods. This is a major reason why a single anatomic test cannot fully characterize coronary health.

Heart Failure: Reduced and Preserved Ejection Fraction

Heart failure is a clinical syndrome in which symptoms and signs arise from structural or functional cardiac abnormality, supported by elevated filling pressures or natriuretic peptide evidence. It is not synonymous with low ejection fraction.

9.1 HFrEF

Heart failure with reduced ejection fraction often results from ischemic injury, dilated cardiomyopathy, toxic injury, myocarditis, genetic disease, persistent tachyarrhythmia, valve disease, or pressure/volume overload. Reduced contractility and dilation lower forward efficiency and increase wall stress. Neurohormonal responses initially preserve pressure and perfusion but promote sodium retention, vasoconstriction, remodeling, arrhythmia, and disease progression.

Modern therapy targets these maladaptive pathways. The major foundational medication classes include renin-angiotensin system/neprilysin modulation, evidence-based beta blockade, mineralocorticoid receptor antagonism, and SGLT2 inhibition, with diuretics for congestion and additional therapies for selected patients (41–43).

9.2 HFpEF

Heart failure with preserved ejection fraction is a heterogeneous syndrome involving impaired relaxation, increased chamber and vascular stiffness, microvascular inflammation, obesity, hypertension, diabetes, kidney disease, atrial dysfunction, chronotropic incompetence, pulmonary hypertension, and skeletal-muscle limitations. Resting EF can be normal while filling pressure rises markedly with exercise.

Diagnosis often requires integration of symptoms, natriuretic peptides, echocardiography, left-atrial and ventricular structure, diastolic indices, exercise testing, and sometimes invasive exercise hemodynamics. Obesity can lower natriuretic peptide concentrations, and atrial fibrillation can raise them, complicating thresholds. SGLT2 inhibitors reduce heart-failure events across a broad EF range (13,44).

9.3 Congestion versus volume

Congestion is elevated filling pressure and venous pressure; total body fluid may be high, normal, or occasionally low. Weight, edema, jugular venous pressure, lung ultrasound, natriuretic peptides, thoracic impedance, pulmonary artery pressure, renal function, and symptoms each provide partial information. Implantable pulmonary artery pressure monitoring can detect hemodynamic changes before overt symptoms in selected patients, but requires an effective clinical response system (45).

9.4 What longitudinal sensors may detect

A multimodal system may detect rising resting heart rate, lower activity, altered sleep, increased respiratory rate, reduced HRV, weight change, PPG morphology changes, edema-related impedance changes, or acoustic changes. These are nonspecific. A useful algorithm should estimate **deviation from personal baseline with uncertainty**, then prompt confirmatory measurement rather than declare heart failure.

Arrhythmias and Sudden Cardiac Death

Arrhythmias arise when impulse formation, conduction, refractoriness, or tissue structure creates abnormal activation. The same rhythm label can have different mechanisms and risks in different hearts.

10.1 Atrial fibrillation

Atrial fibrillation involves disorganized atrial activation with irregular ventricular response. Risk rises with age, hypertension, obesity, sleep apnea, diabetes, valve disease, atrial enlargement, alcohol exposure, thyroid disease, and genetic susceptibility. Consequences include symptoms, tachycardia-mediated dysfunction, heart failure, and thromboembolism.

Detection depends strongly on monitoring duration and population. Consumer PPG can identify pulse irregularity; single-lead ECG can document rhythm; a 12-lead ECG provides additional morphology. Apple and Fitbit population studies demonstrated that large-scale wearable notification is feasible, but enrolled device owners, required confirmatory pathways, and did not imply that every irregular pulse is AF or that absence of notification excludes AF (16,17,46).

10.2 Bradyarrhythmias and conduction disease

Sinus-node dysfunction, atrioventricular block, bundle-branch disease, medication effects, ischemia, electrolyte disorders, and sleep-related pauses can produce bradycardia. Symptoms and context matter. A device must distinguish true pauses from electrode loss, saturation, clipped QRS complexes, and algorithmic missed beats.

10.3 Ventricular arrhythmias

Premature ventricular complexes range from benign to markers or causes of cardiomyopathy depending on burden, morphology, coupling, symptoms, and structural disease. Sustained ventricular tachycardia or fibrillation can cause sudden cardiac death. Risk assessment incorporates ejection fraction, scar, genetics, symptoms, prior events, imaging, and sometimes electrophysiology - not just ectopy counts.

10.4 Long QT, Brugada, and inherited electrical disease

Channelopathies can create malignant arrhythmia risk with structurally normal hearts. Phenotype may be intermittent and modified by drugs, fever, electrolytes, sex, and autonomic state. Automated ECG interpretation can flag patterns but requires expert confirmation. Consumer devices with one lead may not view the diagnostic lead orientation.

Valve, Myocardial, Congenital, Aortic, and Peripheral Vascular Disease

11.1 Valve disease

Stenotic valves create pressure overload; regurgitant valves create volume overload. Severity is evaluated by integrated anatomy, velocity, gradients, valve area, chamber response, flow state, and symptoms. Doppler pressure gradients use the simplified Bernoulli relation,

$$\Delta P \approx 4v^2,$$

where v is peak jet velocity in m/s. The assumptions matter: velocity alignment, proximal velocity, pressure recovery, flow rate, and measurement quality can alter results.

11.2 Cardiomyopathies

Cardiomyopathies include hypertrophic, dilated, arrhythmogenic, restrictive, noncompaction, and other phenotypes. Causes include sarcomeric and cytoskeletal variants, infiltrative disease, inflammation, toxins, pregnancy, tachycardia, endocrine disease, and unknown mechanisms. Genotype does not always predict phenotype; family screening combines history, ECG, imaging, and genetics.

Strain imaging, CMR tissue characterization, and rhythm monitoring can reveal subclinical disease, but each has false positives and vendor or protocol dependence. A wearable cannot rule out cardiomyopathy from normal rate and rhythm.

11.3 Congenital heart disease

Congenital lesions alter anatomy, pressure, flow, saturation, and chamber work in diverse ways. Adults with repaired lesions may retain scars, residual shunts, valve disease, pulmonary hypertension, arrhythmia risk, or ventricular dysfunction. Standard adult reference ranges and algorithms may not apply.

11.4 Aortic disease

Aortic aneurysm and dissection reflect inherited connective-tissue disorders, bicuspid aortic valve, hypertension, smoking, inflammation, atherosclerosis, and degenerative change. Diameter, growth rate, body size, family history, syndrome, and segment determine risk. Imaging reproducibility and consistent measurement planes are critical (47).

11.5 Peripheral artery disease

Lower-extremity PAD is systemic atherosclerosis associated with limb symptoms, impaired healing, amputation risk, and high cardiovascular risk. Resting ankle-brachial index, exercise ABI, toe pressures, pulse-volume recordings, Doppler waveforms, duplex ultrasound, CT/MR angiography, and invasive angiography provide progressively detailed information. Medial arterial calcification can make ankle arteries incompressible, especially in diabetes and kidney disease; toe pressure may be more informative (48,49).

11.6 Venous thromboembolism and pulmonary vascular disease

Deep venous thrombosis and pulmonary embolism involve venous stasis, endothelial injury, hypercoagulability, and provoking factors. They are cardiovascular diseases but differ from atherosclerosis. Diagnosis combines pretest probability, D-dimer in selected patients, ultrasound, CT pulmonary angiography, ventilation-perfusion imaging, and hemodynamic assessment. Wearable heart rate, oxygen saturation, or respiratory rate cannot rule them in or out.

Part III

Determinants of Cardiovascular Health

Genetics, Family History, and Epigenetic Regulation

12.1 What heritability means - and does not mean

Heritability is the fraction of observed variation in a trait within a defined population that is statistically attributable to inherited variation under that population's environmental conditions. It is not the fraction of an individual's disease "caused by genes," and it is not fixed across populations or time. A highly heritable trait can remain modifiable; a low-heritability trait can still be biologically important.

Family and twin studies often place coronary artery disease heritability in a broad range of approximately 40% to 60%, but estimates depend on phenotype, age, sex, ancestry, ascertainment, and shared environment. Heritability of intermediate traits varies: lipoprotein(a) concentration is strongly genetically determined; LDL-C, blood pressure, obesity, diabetes, and smoking behavior have both genetic and environmental components; physical activity and diet are socially constrained and behaviorally mediated.

Family history remains useful because it captures known variants, unknown variants, shared behavior, shared exposures, access to care, and sometimes early subclinical disease. A premature event in a first-degree relative is more informative than a late event in a distant relative. Family history should prompt earlier and more complete risk assessment, not fatalism.

12.2 Monogenic disease

Rare variants can have large effects. Examples include:

- **Familial hypercholesterolemia:** pathogenic variants affecting LDL receptor function, apolipoprotein B binding, PCSK9, or related pathways cause high lifelong LDL exposure and early atherosclerosis.
- **Inherited cardiomyopathies:** sarcomeric, cytoskeletal, desmosomal, nuclear-envelope, mitochondrial, and other variants can cause hypertrophic, dilated, arrhythmogenic, or restrictive phenotypes.
- **Channelopathies:** variants in sodium, potassium, calcium, and accessory proteins can cause long-QT syndromes, Brugada syndrome, catecholaminergic polymorphic ventricular tachycardia, and related disorders.
- **Aortopathies:** variants affecting fibrillin, TGF-beta signaling, smooth-muscle contractile proteins, and extracellular matrix can predispose to aneurysm and dissection.
- **Thrombophilias:** selected coagulation variants increase venous thrombosis risk more than atherosclerotic risk.

A pathogenic variant is interpreted with phenotype, segregation, penetrance, ancestry, age, and variant classification. "Variant of uncertain significance" is not a diagnosis and should not drive irreversible decisions.

12.3 Polygenic risk

Common cardiovascular disease reflects many variants of small effect. Polygenic risk scores aggregate weighted alleles into a relative-risk distribution. Large studies demonstrate that a small fraction of the population can have polygenic coronary risk comparable to some monogenic disorders (7,8). However, clinical implementation has unresolved issues:

- performance differs across ancestry groups because training datasets and linkage structures differ;
- scores are relative to a reference population and can shift with model version;
- calibration, absolute risk, and treatment benefit matter more than percentile alone;
- a score does not capture all rare variants, family history, environment, or current subclinical disease;
- disclosure can change behavior but can also create anxiety or false reassurance.

The most credible use is as one risk-enhancing input within a broader profile, especially when traditional risk is uncertain. It is not a replacement for LDL, blood pressure, diabetes, smoking, kidney function, family history, or coronary calcium.

12.4 Gene-environment interaction

Genes can alter the effect of exposure, and exposure can reveal or suppress genetic risk. Familial hypercholesterolemia creates high cumulative LDL exposure from birth, but smoking, pressure, diabetes, and treatment timing strongly modify event age. Sarcomeric variants may remain clinically silent for years, then become manifest with age or loading. Alcohol can trigger atrial fibrillation in susceptible people. Fever can reveal a Brugada pattern. Pregnancy can unmask cardiomyopathy or vascular vulnerability.

The NEJM analysis by Khera and colleagues found that favorable lifestyle was associated with substantially lower coronary risk across genetic-risk strata, including nearly 50% lower relative risk among those at high genetic risk (9). This is observational within cohort data rather than random allocation of lifestyle, but it directly refutes the idea that polygenic risk makes prevention irrelevant.

12.5 Epigenetics and developmental programming

DNA methylation, histone modification, chromatin organization, and noncoding RNA respond to age, nutrition, smoking, inflammation, and metabolic state. Some marks are consequences rather than causes, and tissue specificity complicates blood-based assays. Maternal health, placental function, fetal growth, childhood adversity, and early nutrition can influence later pressure, adiposity, insulin sensitivity, and vascular structure. The developmental-origins framework is plausible and supported by multiple lines of evidence, but individual prediction from epigenetic clocks or commercial methylation panels is not yet sufficiently validated for routine cardiovascular decision-making.

Genes set a distribution of susceptibility and can occasionally dominate a phenotype, but common cardiovascular disease is usually the product of inherited susceptibility interacting with decades of exposure. Genetic information is most useful when it changes screening, treatment intensity, family testing, or interpretation - not when it is converted into a deterministic "heart age."

Lipids, Lipoproteins, and Atherosclerotic Exposure

13.1 LDL-C, non-HDL-C, apoB, and particle number

LDL-C is the mass of cholesterol carried in LDL particles. It is highly useful, standardized, and the principal treatment metric in many settings. Non-HDL-C subtracts HDL-C from total cholesterol and captures cholesterol carried by all apoB particles. ApoB approximates the number of atherogenic particles. These measurements usually agree, but discordance is common in hypertriglyceridemia, diabetes, obesity, and metabolic syndrome.

ApoB can clarify risk when LDL-C appears acceptable but particle number is high. Non-HDL-C is inexpensive and available from a standard lipid panel. Direct LDL assays can have method-specific limitations; calculated LDL becomes less reliable at high triglycerides or very low LDL, though newer equations improve estimation.

13.2 Lipoprotein(a)

Lp(a) is an LDL-like particle covalently attached to apolipoprotein(a). Concentration is largely genetic, varies substantially across individuals and ancestry groups, and is relatively stable after childhood except in certain inflammatory, kidney, liver, or hormonal states. It is associated with atherosclerotic cardiovascular disease and calcific aortic valve disease (23).

Assays may report mg/dL mass or nmol/L particle concentration; conversion with one universal factor is unreliable because apo(a) isoform size varies. Current guidance recommends at least once-in-lifetime measurement for broad risk assessment and more targeted testing in premature disease or family history (12).

13.3 Triglyceride-rich remnants

Triglycerides are carried in chylomicrons, VLDL, and their remnants. Large chylomicrons may be too large to enter the intima efficiently, but remnant particles can be atherogenic. Severe hypertriglyceridemia creates pancreatitis risk. Moderate elevation often indicates insulin resistance, obesity, alcohol exposure, diet, diabetes, kidney disease, hypothyroidism, or medication effects.

The cardiovascular benefit of triglyceride-lowering therapy depends on agent and population; lowering the laboratory number is not sufficient proof. Treatment begins with secondary causes, weight, diet quality, activity, glycemic control, alcohol, and apoB-based risk management.

13.4 HDL: marker, mechanism, and failed simplification

Low HDL-C predicts risk in observational cohorts, but pharmacologically raising HDL-C has not reliably reduced events. HDL particles are heterogeneous and participate in cholesterol transport, inflammation, oxidation, and endothelial function. HDL-C concentration is therefore a risk marker but not a simple causal treatment target. Extremely high HDL-C is not necessarily protective.

13.5 Cumulative burden

Atherosclerotic injury integrates particle concentration over time. The same LDL-C measured at age 25 and age 75 implies different accumulated exposure. Genetic studies support larger lifetime risk reduction per unit LDL difference when exposure is lower from early life than when treatment begins late. This is why contemporary guidance increasingly emphasizes childhood, young-adult, and lifetime risk (4,12).

Glycemia, Adiposity, Insulin Resistance, and Kidney Function

14.1 Diabetes and vascular injury

Diabetes increases atherosclerotic disease, heart failure, kidney disease, peripheral vascular disease, and microvascular complications. Mechanisms include glycation, oxidative stress, inflammation, endothelial dysfunction, dyslipidemia, prothrombotic state, autonomic neuropathy, and renal injury. Risk varies with duration, age, blood pressure, lipids, kidney function, smoking, and prior disease.

HbA1c summarizes average glycemic exposure over approximately two to three months but is affected by red-cell lifespan, anemia, hemoglobin variants, transfusion, pregnancy, and kidney disease. Fasting glucose, oral glucose tolerance, and continuous glucose monitoring answer different questions. Glycemic variability is biologically plausible but its incremental cardiovascular value beyond mean exposure and clinical context is still being refined.

14.2 The lesson from outcome trials

Glucose lowering is not a single mechanism. Some therapies improve glycemia without reducing cardiovascular events; others have organ-specific benefits beyond glucose. SGLT2 inhibitors reduce heart-failure and kidney events across multiple populations, including some without diabetes (42–44,50–52). GLP-1 receptor agonists reduce atherosclerotic outcomes in diabetes, and SELECT showed cardiovascular benefit with semaglutide in adults with overweight or obesity and established cardiovascular disease but without diabetes (53,54).

This evidence cautions against treating HbA1c as the only endpoint. Cardiovascular and renal outcomes depend on blood pressure, weight, volume, inflammation, kidney hemodynamics, and drug class.

14.3 Obesity and ectopic fat

Body mass index is a scalable screening measure but does not distinguish fat, muscle, distribution, or ectopic deposits. Waist circumference, waist-to-height ratio, imaging, and metabolic markers provide complementary information. Visceral and ectopic fat are linked to insulin resistance, inflammation, hepatic lipid production, sleep apnea, HFpEF, and atrial fibrillation.

Obesity is not merely a behavior label. It reflects genetics, neuroendocrine appetite regulation, food environment, sleep, medications, mental health, socioeconomic conditions, and energy balance. The 2026 CKM guideline explicitly treats obesity as a biological and clinical disease linked to cardiovascular and kidney outcomes (13).

Intentional weight loss improves pressure, glycemia, triglycerides, sleep apnea, mobility, and symptoms. Event reduction depends on method and population. Look AHEAD improved many risk factors but did not significantly reduce its primary cardiovascular endpoint, whereas SELECT demonstrated event reduction with semaglutide in a high-risk population (54,55). Bariatric/metabolic surgery has strong observational and randomized metabolic evidence, but procedure choice and long-term risk require specialist care.

14.4 Chronic kidney disease

Kidney disease is both a cardiovascular risk enhancer and a mechanistic driver. Reduced filtration, albuminuria, sodium retention, anemia, mineral-bone disorder, inflammation, uremic toxins, sympathetic activation, and vascular calcification increase risk. Estimated GFR and urine albumin-creatinine ratio provide complementary information: albuminuria can identify vascular and glomerular injury even when filtration is preserved.

A cardiovascular profile that omits kidney measures is incomplete. The PREVENT equations and 2026 CKM framework include kidney and metabolic information because they materially alter risk and treatment benefit (13,19).

Nutrition: Patterns, Nutrients, Processing, and Evidence

Nutrition affects cardiovascular health through lipoproteins, blood pressure, glycemia, body weight, microbiome metabolites, inflammation, kidney load, and displacement of healthier or less healthy foods. The evidence is strongest for overall dietary patterns and replacement effects, not isolated nutrient slogans.

15.1 Dietary patterns

The DASH trial demonstrated that a diet rich in fruits, vegetables, and low-fat dairy with reduced saturated fat lowered blood pressure even without weight loss. DASH-Sodium showed additional pressure reduction with lower sodium, with large effects among some hypertensive participants (56,57).

Mediterranean-style patterns emphasize vegetables, fruit, legumes, whole grains, nuts, olive oil or other unsaturated fats, fish, and limited processed meat and refined foods. PREDIMED, after republication addressing randomization irregularities, found fewer major cardiovascular events in high-risk adults assigned to Mediterranean diets supplemented with extra-virgin olive oil or nuts. CORDIOPREV supported a Mediterranean pattern for secondary prevention compared with a low-fat diet (58,59).

The current AHA framework describes nine features: appropriate energy balance; varied vegetables and fruits; mostly whole grains; healthy protein sources; unsaturated rather than saturated fats; minimally processed rather than ultraprocessed foods; minimal added sugars; lower sodium; and no initiation of alcohol for health (14).

15.2 Saturated and unsaturated fats

Replacing saturated fat with polyunsaturated fat lowers LDL-C and cardiovascular risk. Replacing saturated fat with refined carbohydrate may lower LDL-C modestly while raising triglycerides and not improving outcomes. The replacement nutrient is therefore central. Trans fats increase risk and have been removed from many food supplies through regulation.

Unsaturated fats from nuts, seeds, fish, and plant oils can improve lipid patterns when they replace saturated fats or refined starches. This is not a license for unlimited energy intake; food matrix and total pattern matter.

15.3 Sodium and potassium

Sodium raises pressure in a dose-responsive manner on average, with wide individual variation. Major sources are often processed and restaurant foods rather than salt added at the table. Potassium-rich foods can support lower pressure and vascular health, but potassium supplementation or salt substitutes can be dangerous in kidney disease or with medications that impair potassium excretion.

A device-based nutrition program cannot infer sodium intake accurately from blood pressure or sweat sodium alone. Sweat concentration depends on rate, acclimatization, temperature, location, and genetics. Repeated weighed diet records, urinary sodium collections, and contextual measures answer different questions.

15.4 Fiber, whole grains, fruits, vegetables, legumes, and nuts

These foods provide fiber, potassium, magnesium, unsaturated fats, phytochemicals, and lower energy density. Soluble fiber can lower LDL-C. Whole grains displace refined grains and improve glycemic response and satiety. Legumes and nuts can replace processed meat or refined snacks. The causal contribution of one phytochemical is rarely separable from the food pattern.

15.5 Fish and omega-3 fatty acids

Fish consumption is part of many heart-healthy patterns. Supplement trials are heterogeneous because dose, formulation, baseline intake, triglycerides, placebo oils, and populations differ. High-dose purified eicosapentaenoic acid reduced events in a selected high-risk, statin-treated population, while several mixed EPA/DHA trials were neutral. Over-the-counter products are not equivalent to trial formulations, and high doses can increase atrial fibrillation risk.

15.6 Added sugar and sugar-sweetened beverages

Added sugars contribute to energy excess, weight gain, triglycerides, fatty liver, and glycemic load, especially in beverages that provide weak satiety. The relevant contrast is what replaces them: water, unsweetened beverages, whole fruit, or another caloric food.

15.7 Ultraprocessed foods

Ultraprocessed-food classifications capture industrial formulations, additives, energy density, texture, palatability, and displacement of minimally processed foods. Prospective studies consistently associate higher intake with cardiovascular and metabolic outcomes, and controlled feeding shows higher spontaneous energy intake under some ultraprocessed diets. However, classification groups diverse products and residual confounding remains. The practical guidance to minimize ultraprocessed foods is reasonable, but a single processing label should not override nutrient quality and context (14).

15.8 Supplements

Routine vitamin, antioxidant, or multivitamin supplementation has not reproduced the cardiovascular benefits of healthy foods. Some supplements interact with anticoagulants, alter pressure or rhythm, contain pharmacologically active contaminants, or cause liver/kidney injury. Correcting a documented deficiency is different from using a supplement as a cardiovascular preventive drug.

Nutrition evidence hierarchy

The highest-confidence recommendation is a sustainable food pattern that improves LDL/apoB exposure, pressure, glycemia, weight trajectory, and dietary quality. Single nutrients matter, but their effect depends on the food source and what they replace. Short-term biomarker changes should not automatically be translated into long-term event claims.

Physical Activity, Fitness, Sedentary Time, and Exercise Response

16.1 Activity versus fitness

Physical activity is behavior; cardiorespiratory fitness is an integrated capacity influenced by activity, genetics, age, heart, lungs, blood, muscle, and motivation. Fitness is one of the strongest predictors of morbidity and mortality, but it is not measured accurately by step count alone (60).

Aerobic activity improves blood pressure, insulin sensitivity, endothelial function, autonomic balance, lipid metabolism, body composition, mood, sleep, and exercise capacity. Resistance training preserves muscle and function, improves glucose disposal, and supports healthy aging. The WHO recommends at least 150 to 300 minutes per week of moderate aerobic activity or 75 to 150 minutes vigorous activity, plus muscle strengthening, while emphasizing that some activity is better than none (61).

16.2 Dose-response

The largest relative benefit occurs when moving from very low activity to modest activity. Benefits continue at higher volumes with diminishing returns. Step-count cohorts show a graded association with mortality, but step thresholds are not universal prescriptions; cadence, intensity, age, and baseline health matter (62).

Sedentary time has independent associations, especially when prolonged and not offset by moderate-to-vigorous activity. Breaking up sitting can improve postprandial glucose and vascular function. Wearables should report distributions, sustained bouts, and context rather than only daily totals.

16.3 Exercise as a physiologic stress test

Exercise raises heart rate, stroke volume, systolic pressure, oxygen extraction, and coronary flow. Healthy systems increase output while maintaining appropriate pressure and rhythm. Abnormal responses include ischemic symptoms or ECG changes, chronotropic incompetence, excessive or blunted pressure response, arrhythmias, desaturation, ventilatory inefficiency, or impaired oxygen uptake.

Cardiopulmonary exercise testing measures breath-by-breath oxygen uptake and carbon dioxide output and can distinguish cardiac, pulmonary, circulatory, muscular, and effort limitations more effectively than a wearable estimate of VO₂ max. Wearable estimates are model-based and depend on movement, heart rate, demographics, terrain, temperature, and calibration.

16.4 Risk and special populations

Exercise is generally protective, but vigorous exertion transiently increases event risk in susceptible, unaccustomed individuals. Structural heart disease, active myocarditis, uncontrolled arrhythmia, severe valve disease, acute coronary syndrome, or decompensated heart failure require individualized guidance. The engineering system should identify symptoms and stop criteria; it should not encourage users to “push through” chest pain, syncope, or severe breathlessness.

Tobacco, Nicotine, Alcohol, and Other Exposures

17.1 Combustible tobacco

Smoking promotes endothelial dysfunction, inflammation, oxidation, platelet activation, thrombosis, sympathetic stimulation, carbon monoxide exposure, adverse lipoproteins, aneurysm, and peripheral disease. There is no safe level of combustible tobacco exposure. Cardiovascular risk falls after cessation, with substantial benefit beginning quickly and accumulating over years (63).

Secondhand smoke also increases risk. Biomarkers such as cotinine can validate exposure but do not capture all combustion products or cumulative injury.

17.2 Electronic cigarettes and nicotine

Electronic cigarettes generally expose users to fewer combustion products than cigarettes but are not harmless. Aerosols can contain nicotine, ultrafine particles, aldehydes, metals, and flavoring chemicals. Nicotine raises heart rate and pressure and may trigger symptoms or arrhythmias in susceptible users. Long-term cardiovascular outcome evidence is less mature than for cigarettes. For a smoker who completely switches, harm may be lower than continued smoking; dual use can preserve substantial risk. The best cardiovascular endpoint is complete cessation of combustible tobacco and ideally nicotine dependence.

17.3 Alcohol

Older observational studies often showed J-shaped associations, but abstainer bias, socioeconomic confounding, reverse causation, beverage pattern, and underreporting complicate interpretation. Mendelian-randomization analyses do not support starting alcohol for cardiovascular protection (64). Alcohol raises blood pressure, triglycerides, atrial fibrillation risk, cardiomyopathy risk at high exposure, sleep disruption, and injury risk. Binge patterns are particularly harmful.

Current AHA dietary guidance: if a person does not drink, do not start for health; if they drink, limit intake (14). A wearable can detect some physiologic response - higher nighttime heart rate, reduced HRV, altered sleep - but cannot quantify safe exposure.

17.4 Stimulants, cannabis, cocaine, and methamphetamine

Cocaine and methamphetamine can cause severe hypertension, vasospasm, thrombosis, myocardial infarction, arrhythmia, cardiomyopathy, aortic dissection, and stroke. Cannabis data are complicated by product variability, route, dose, co-use, and confounding; acute tachycardia and autonomic effects are common, and cardiovascular risk signals warrant caution. Energy drinks combine caffeine with other stimulants and may affect pressure, rhythm, and sleep, especially at high doses or with exercise, alcohol, or underlying disease.

Sleep, Circadian Biology, Stress, and Mental Health

18.1 Sleep duration and regularity

Short, long, fragmented, and irregular sleep are associated with obesity, diabetes, hypertension, inflammation, autonomic imbalance, and cardiovascular events. Sleep became a formal component of Life's Essential 8 because it is both a behavior and a physiological regulator (65,66).

Consumer sleep staging is inferred from movement, PPG, and sometimes temperature or oxygen saturation. It can estimate patterns but does not equal polysomnography. Device algorithms and firmware changes can change stage labels without a biological change. For cardiovascular profiling, sleep duration, consistency, resting heart rate, respiratory rate, oxygen-desaturation patterns, and symptom context may be more robust than exact consumer stage percentages.

18.2 Obstructive sleep apnea

Obstructive sleep apnea produces recurrent airway collapse, hypoxemia, arousal, intrathoracic pressure swings, sympathetic surges, and sleep fragmentation. It is associated with hypertension, atrial fibrillation, stroke, pulmonary hypertension, and heart failure (67).

A wrist or ring can screen for repetitive desaturation or pulse patterns but cannot fully classify apnea mechanism, arousal, sleep time, or respiratory effort. Home sleep apnea testing and polysomnography use validated respiratory channels and scoring criteria.

18.3 Stress pathways

Acute stress activates sympathetic and hypothalamic-pituitary-adrenal systems, raising heart rate, pressure, vasoconstriction, platelet activity, and metabolic demand. Chronic stress can affect sleep, diet, activity, smoking, medication adherence, inflammation, and social connection. Severe emotional or physical stress can trigger takotsubo syndrome in susceptible individuals.

Psychosocial stress was strongly associated with myocardial infarction in INTERHEART, and job-strain meta-analyses show modest increased coronary risk (10,68). Yet stress is difficult to quantify. Self-report, cortisol, HRV, electrodermal activity, sleep, and context measure different dimensions. A low HRV value is not a direct stress meter.

18.4 Depression, anxiety, and social isolation

Depression and anxiety are associated with cardiovascular risk, symptoms, adherence, autonomic and inflammatory changes, and outcomes after disease. Social isolation and loneliness can affect behavior and physiology. Treatment improves well-being and may improve adherence and function, but cardiovascular event reduction is not guaranteed for every psychological intervention (69).

A humane monitoring system should avoid converting normal emotional variation into alarms. It should give users control over interpretation and privacy, avoid punitive scoring, and connect concerning trends to appropriate human support.

Air Pollution, Noise, Temperature, and the Built Environment

19.1 Air pollution

Fine particulate matter can enter the lungs and trigger oxidative stress, inflammation, autonomic imbalance, endothelial dysfunction, thrombosis, and vascular effects. Short-term exposure can trigger events in susceptible people; long-term exposure contributes to atherosclerosis and cardiopulmonary risk. The biological effect depends on particle size, composition, source, co-pollutants, and vulnerability (70).

Personal exposure differs from a city monitor because of indoor sources, filtration, occupation, commuting, cooking, and time-activity patterns. Integrating geospatial air-quality data with a wearable can improve exposure estimates, but GPS and home-location data create substantial privacy risk.

19.2 Noise and sleep disruption

Traffic and aircraft noise can disturb sleep and activate stress pathways even when a person reports habituation. Exposure assessment requires frequency, timing, insulation, bedroom orientation, and individual susceptibility. Noise may operate through sleep fragmentation, stress hormones, pressure, and behavior.

19.3 Heat and cold

Heat increases skin blood flow, sweating, dehydration risk, and cardiovascular strain. Cold increases vasoconstriction and pressure. Extreme temperatures are associated with cardiovascular events, especially in older adults and people with heart failure, kidney disease, or limited access to climate control. Medication effects matter: diuretics, vasodilators, beta blockers, and anticholinergic drugs can alter heat response.

19.4 Built environment and access

Walkability, green space, food availability, transport, housing quality, neighborhood safety, and healthcare access shape daily exposure. These are not background variables; they influence activity, diet, pollution, sleep, stress, and treatment. Social determinants can create risk before an individual makes any conscious “lifestyle choice” (13,71).

Sex, Reproductive History, Age, and Population Context

20.1 Biological sex and gendered exposure

Women and men share major risk factors, but presentation, lifetime exposure, vessel size, hormones, pregnancy, care patterns, and symptom recognition differ. Women may have more coronary microvascular dysfunction or nonobstructive ischemic syndromes, while men often develop obstructive coronary disease earlier. Gendered differences in occupation, caregiving, violence, income, and access can modify risk.

20.2 Pregnancy-related risk enhancers

Hypertensive disorders of pregnancy, gestational diabetes, preterm delivery, fetal growth restriction, and pregnancy-associated cardiomyopathy identify higher later cardiovascular risk. Pregnancy is a physiologic stress test that can reveal vascular, metabolic, and myocardial susceptibility. These histories should be incorporated into long-term risk assessment rather than forgotten after delivery.

20.3 Menopause and hormone therapy

Menopause is associated with changes in lipids, body composition, vascular function, and symptoms. Randomized evidence does not support initiating menopausal hormone therapy solely to prevent cardiovascular disease; timing, formulation, route, symptoms, thrombosis risk, and individual context matter.

20.4 Aging

Aging affects arterial stiffness, conduction tissue, autonomic response, renal function, muscle, valve calcification, inflammation, and repair. Chronological age is a powerful risk predictor because it integrates duration of exposure, but it should not be treated as an untreatable mechanism. Older adults can gain substantial absolute benefit from risk reduction, balanced against frailty, falls, polypharmacy, kidney function, life expectancy, and preferences.

20.5 Ancestry and race

Genetic ancestry can influence allele frequencies and some disease mechanisms, but socially defined race also captures structural conditions, discrimination, geography, care, and exposure. Algorithms that use race as a biological correction can encode inequity. Pulse oximetry performance differences across skin pigmentation are a direct device example: optical calibration and validation populations can create clinically meaningful bias (72).

Quantifying Relative Contributions Without False Precision

The desire for one chart assigning fixed percentages to genes, food, exercise, stress, and environment is understandable but scientifically misleading. Several distinct quantities are often confused.

21.1 Five different questions

1. **Heritability:** What fraction of variation in this population is associated with inherited variation?
2. **Relative risk:** How much higher is risk in exposed versus unexposed groups?
3. **Population-attributable fraction:** What proportion of events might be avoided under a counterfactual removal of an exposure, assuming causality and the model?
4. **Individual attributable cause:** Which factors produced one person's event? Usually not uniquely identifiable.
5. **Treatment effect:** How much does a specific intervention change risk in a defined population over a defined time?

These quantities are not interchangeable.

21.2 Why attributable fractions do not add cleanly

Risk factors cluster. Obesity can increase pressure, diabetes, sleep apnea, inflammation, and lipids. Poverty can change food, pollution, stress, sleep, and access. Smoking interacts with lipids and thrombosis. Removing one exposure changes the distribution of others. Population-attributable fractions from multivariable models depend on prevalence, measurement error, reference category, interactions, and model specification; they can sum above 100%.

21.3 A practical weighting framework

For engineering and prevention, use a **causal-confidence x exposure-burden x actionability** matrix rather than fixed percentages.

Factor	Causal confidence	Typical exposure horizon	Practical measurement	Actionability
ApoB/LDL particle exposure	Established	Years to decades	Lipid panel, apoB, Lp(a)	High with lifestyle and medication
Blood pressure	Established	Continuous, cumulative	Validated of-fice/home/ambulatory BP	High
Smoking/combustion	Established	Dose and duration	History, cotinine in selected settings	High but dependence treatment needed
Diabetes/glycemia	Established	Years	HbA1c, glucose, CGM context	High
Kidney disease	Established/risk amplifier	Years	eGFR, urine albumin	Moderate to high; cause-specific
Physical inactivity/low fitness	Established/probable causal	Months to decades	Activity plus exercise capacity	High

Factor	Causal confidence	Typical exposure horizon	Practical measurement	Actionability
Diet pattern	Established for key pathways/patterns	Months to decades	Repeated dietary assessment and biomarkers	Moderate to high
Obesity/ectopic fat	Established/risk amplifier	Years	BMI, waist, metabolic profile, imaging	Moderate to high
Sleep apnea	Established association; treatment effects endpoint-specific	Nightly, chronic	Sleep study/reference testing	High for symptoms and selected outcomes
Stress/depression	Important, partly causal and mediated	Acute and chronic	Self-report plus context; no single biomarker	Moderate; treatment essential for well-being
Air pollution	Established population risk	Acute and chronic	Modeled/personal exposure	Individual control limited; policy important
Polygenic risk	Risk stratifier, mechanism-dependent	Lifelong	Validated ancestry-aware score	Actionable through earlier/intensified prevention

A causal web, not a checklist

Most cardiovascular outcomes emerge from interacting lifelong exposures

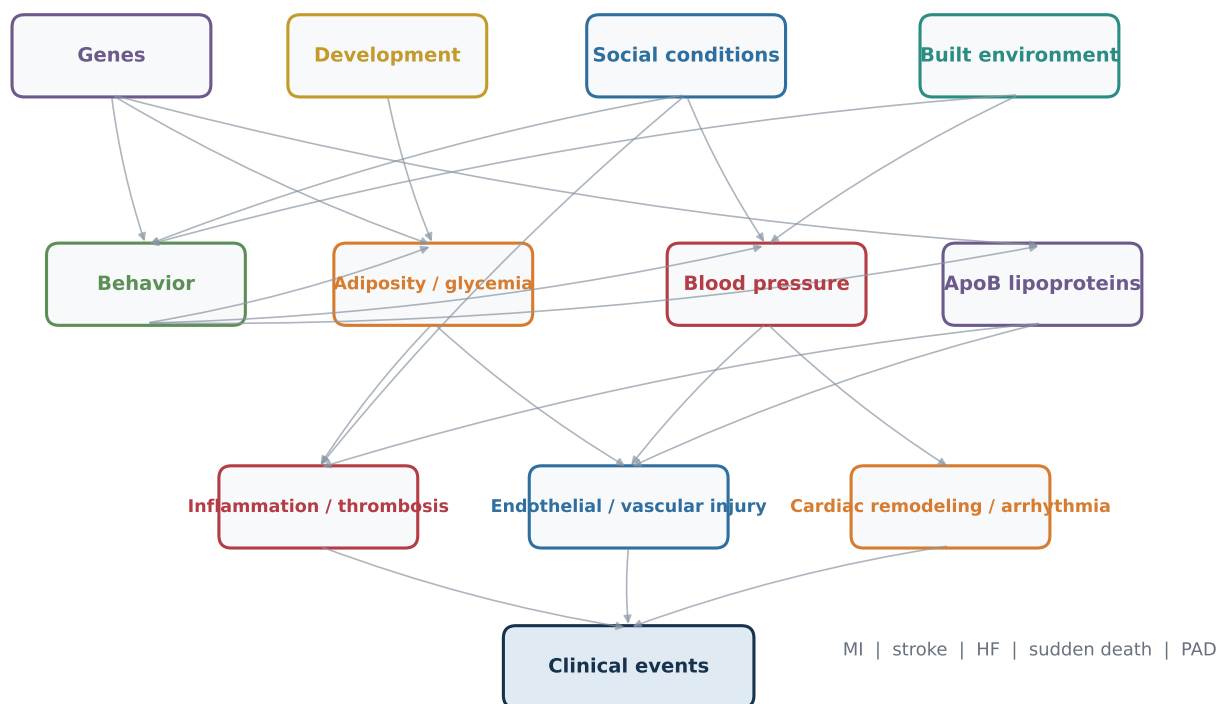


Figure 21.1: Causal web linking inherited susceptibility, social and physical environment, behavior, metabolic/renal state, subclinical injury, triggers, and clinical disease.

21.4 Synthesis of the major estimates

- Twin and family studies commonly estimate CAD heritability around 40% to 60%, but this is population-specific and not an individual destiny.
- INTERHEART attributed more than 90% of first-MI population risk to nine modifiable factors, but it was a case-control study of MI and used model-based counterfactuals (10).
- The Global Cardiovascular Risk Consortium estimated that five major modifiable factors explained approximately 57% of incident CVD in women and 53% in men across studied cohorts; the residual includes measurement error, unmeasured exposures, age, biology, and stochastic events (6).
- High genetic risk and healthy lifestyle coexist; favorable behavior is associated with substantially lower coronary risk at every genetic level (9).

Most defensible answer

Common cardiovascular disease is neither "mostly genetic" nor "mostly lifestyle" in a universal sense. Inherited biology materially shapes susceptibility, but a large fraction of population events is associated with modifiable exposures, and treatment of LDL, pressure, tobacco exposure, diabetes, kidney-metabolic disease, and inactivity has proven benefit. The useful question is not which side wins; it is which causal exposures and latent disease can be measured early enough to change outcome.

Part IV

Evidence, Prevention, and Current Clinical Assessment

Landmark Evidence and What It Established

The table below is selective rather than exhaustive. It emphasizes studies that changed causal understanding, prevention, or measurement. Outcome magnitude should always be interpreted with population, follow-up, comparator, adherence, and absolute risk.

Study	Design / population	Principal contribution
Framingham Heart Study	Longitudinal community cohort	Established multivariable risk-factor framework and long-term prediction.
INTERHEART	International case-control study	Demonstrated global consistency of major modifiable MI risk factors (10).
Global Cardiovascular Risk Consortium	Pooled prospective cohorts	Quantified contributions of five major modifiable factors across regions (6).
DASH	Controlled feeding trial	Demonstrated blood-pressure reduction from a dietary pattern independent of weight loss (56).
DASH-Sodium	Controlled feeding crossover	Demonstrated additive pressure effects of lower sodium and DASH pattern (57).
PREDIMED	Randomized primary-prevention diet trial	Supported Mediterranean dietary patterns in high-risk adults (58).
SPRINT	Randomized BP target trial	Demonstrated benefit of intensive systolic control in selected high-risk adults without diabetes using standardized measurement (35).
ACCORD BP	Randomized BP target trial in diabetes	Highlighted endpoint-specific benefit and tradeoffs; stroke reduction despite neutral primary composite (36).
4S and HPS	Randomized statin trials	Established large clinical benefit of LDL lowering in secondary/high-risk prevention (25,26).
IMPROVE-IT	Ezetimibe plus statin	Showed incremental event reduction from nonstatin LDL lowering (27).
FOURIER / ODYSSEY	PCSK9 inhibitor outcomes	Confirmed benefit at very low achieved LDL in high-risk patients (28,29).
CANTOS	Anti-IL-1 beta trial	Demonstrated a lipid-independent causal inflammatory pathway (31).
DPP	Randomized prevention trial	Showed intensive lifestyle and metformin reduce diabetes onset (73).
Steno-2	Multifactorial diabetes intervention	Demonstrated durable benefit of integrated risk-factor treatment (74).
EMPA-REG and later SGLT2 trials	Cardiovascular/renal outcome trials	Established heart-failure and kidney benefits beyond glucose lowering (42,44,50).
SELECT	Semaglutide in obesity without diabetes	Demonstrated cardiovascular event reduction in established CVD with overweight/obesity (54).
Apple / Fitbit Heart Studies	Very large pragmatic wearable studies	Demonstrated scalable AF screening and the need for confirmatory pathways (16,17).
FAME / DEFINE-FLAIR	Invasive coronary physiology trials	Established physiological lesion assessment beyond visual stenosis (39,40).

Study	Design / population	Principal contribution
CHAMPION	Implantable PA pressure monitoring	Showed benefit of pressure-guided management in selected heart-failure care systems (45).

22.1 How to read trial evidence

A relative-risk reduction can sound dramatic while absolute benefit is small in a low-risk population. Conversely, a modest relative effect can prevent many events in a high-risk population. Key quantities are:

$$ARR = Risk_{control} - Risk_{treatment}$$

$$NNT = \frac{1}{ARR}$$

over the stated time. NNT is not a property of a drug alone; it changes with baseline risk, follow-up, adherence, and outcome definition. Harms require the same analysis.

Surrogate outcomes are useful when strongly validated and mechanistically linked, but not all surrogate changes translate to fewer events. LDL lowering and blood-pressure lowering have robust outcome relationships across drug classes, yet off-target effects still matter. Raising HDL-C, suppressing ectopy, or improving one AI-derived score does not automatically improve clinical outcomes.

22.2 Observational studies and healthy-user effects

Diet, exercise, sleep, alcohol, stress, and supplement studies are especially vulnerable to confounding. People who adhere to one healthy behavior often differ in education, income, healthcare, smoking, and other behaviors. Repeated measures, negative controls, natural experiments, Mendelian randomization, and trials help, but no method is perfect. The packet therefore describes evidence convergence rather than treating one study design as infallible.

Primordial, Primary, and Secondary Prevention

23.1 Primordial prevention

Primordial prevention prevents the development of risk factors: healthy food environments, tobacco control, physical activity, sleep, clean air, healthy pregnancy, childhood prevention, and equitable access. It can produce more population benefit than treating established disease one person at a time, but benefits are delayed and distributed across systems.

23.2 Primary prevention

Primary prevention targets people without a recognized cardiovascular event. The core process is:

1. confirm measurements and identify secondary causes;
2. estimate short- and long-term risk;
3. identify risk enhancers and family history;
4. assess subclinical disease when it will change a decision;
5. implement lifestyle and, when indicated, pharmacologic therapy;
6. monitor efficacy, adverse effects, adherence, and access.

PREVENT estimates 10- and 30-year risk and incorporates cardiovascular, kidney, and metabolic variables (19). Coronary artery calcium can reclassify selected borderline or intermediate decisions, but zero calcium does not erase smoking, diabetes, severe hypercholesterolemia, strong family history, or noncalcified plaque.

23.3 Secondary prevention

After myocardial infarction, stroke, symptomatic PAD, or established atherosclerotic disease, baseline risk is high. Therapy generally includes intensive lipid lowering, blood-pressure control, tobacco cessation, antithrombotic therapy when indicated, physical rehabilitation, diabetes/CKM management, and disease-specific procedures. The 2026 dyslipidemia guideline recommends very low LDL-C goals for very-high-risk secondary prevention, reflecting cumulative trial evidence (12).

23.4 Cardiovascular-kidney-metabolic staging

The 2026 CKM guideline unifies adiposity, metabolic risk, kidney disease, and clinical cardiovascular disease. The engineering implication is important: a “heart monitor” should not be designed as an isolated ECG gadget. Weight trajectory, waist, pressure, glucose, kidney function, sleep, and activity often reveal risk earlier than rhythm abnormality (13).

Current Risk Assessment and a Complete Baseline Profile

24.1 History

A high-quality cardiovascular history includes:

- symptoms: exertional chest discomfort, dyspnea, orthopnea, edema, palpitations, syncope, claudication, neurologic episodes;
- time course and triggers;
- prior cardiovascular, kidney, metabolic, inflammatory, pulmonary, or sleep disease;
- family history including age and phenotype;
- pregnancy history and reproductive risk markers;
- tobacco, nicotine, alcohol, stimulants, and other substances;
- diet, activity, occupation, sleep, stress, finances, housing, and access;
- medication and supplement list, adherence, side effects, and affordability.

24.2 Examination

Core measurements include standardized bilateral-arm pressure when appropriate, heart rate and rhythm, orthostatic response when indicated, body size and waist, oxygen saturation in context, pulses, heart sounds, jugular venous pressure, edema, lung examination, and vascular findings. Examination is not obsolete: it identifies device placement issues, perfusion, congestion, and contradictions requiring confirmation.

24.3 Baseline laboratory panel

A general profile commonly includes:

- lipid panel with non-HDL-C; apoB when discordance or risk warrants;
- Lp(a) at least once in adulthood according to current guidance;
- fasting glucose and/or HbA1c;
- creatinine/eGFR and urine albumin-creatinine ratio;
- electrolytes and liver tests when relevant to disease or therapy;
- blood count for anemia;
- thyroid studies when rhythm, lipids, or symptoms suggest;
- high-sensitivity troponin or natriuretic peptide only for appropriate clinical questions, not indiscriminate screening.

High-sensitivity CRP can refine risk in selected settings but is nonspecific. Novel omics panels may improve prediction in research, yet incremental utility, calibration, cost, and actionability must be demonstrated.

24.4 Baseline instruments

A practical noninvasive baseline combines:

1. validated upper-arm office and home blood pressure;
2. resting 12-lead ECG;
3. targeted ambulatory rhythm monitoring if symptoms or risk justify;
4. echocardiography when structural disease, murmur, heart failure, or abnormal ECG suggests;

5. vascular assessment such as ABI when PAD risk or symptoms exist;
6. exercise testing or CPET when functional capacity or exertional symptoms require explanation;
7. coronary calcium or CT angiography only when it changes a clinical decision.

What Is Known, What Is Probable, and What Remains Unsettled

25.1 High-confidence conclusions

- ApoB-containing lipoprotein exposure is causal in atherosclerosis; lowering LDL/apoB exposure reduces events.
- Elevated blood pressure causes vascular, cardiac, kidney, and brain injury; accurate measurement and treatment reduce events.
- Combustible tobacco causes cardiovascular disease; cessation reduces risk.
- Diabetes, kidney disease, and adverse adiposity substantially amplify risk; integrated treatment improves outcomes.
- Physical activity and cardiorespiratory fitness are strongly protective; inactivity is harmful.
- Heart-healthy dietary patterns improve major risk pathways and reduce events in selected trials.
- Sleep apnea is an important comorbidity; sleep duration and regularity are meaningful health dimensions.
- Air pollution and social conditions materially affect population cardiovascular risk.
- No single wearable or blood test provides a complete cardiovascular health score.

25.2 Probable or context-dependent conclusions

- HRV, resting heart rate, respiratory rate, temperature, activity, and sleep deviations can improve early-warning systems when interpreted against personal baseline.
- Coronary calcium is a powerful risk reclassifier in selected asymptomatic adults, but its value depends on pretest risk and treatment decision.
- Multi-sensor wearable models can detect AF and selected decompensation patterns, but outcome benefit depends on confirmatory care and false-positive management.
- Some inflammatory markers and pathways improve risk stratification, but broad anti-inflammatory treatment is not universally appropriate.
- Ectopic fat and imaging-based body composition may improve prediction beyond BMI, but availability and actionability vary.

25.3 Unsettled or frequently overclaimed

- Calibration-free cuffless blood pressure from PPG/PTT in free living.
- Consumer “arterial age,” “vascular age,” or “stress” scores without transparent reference and validation.
- Detecting coronary stenosis or heart failure reliably from a single-lead ECG in unselected populations.
- Routine polygenic risk screening across ancestries without demonstrated decision impact.
- Long-term cardiovascular outcome benefit from consumer supplements, cold exposure, saunas, intermittent fasting, or other trends independent of total pattern and selection effects.
- Generalizing an AI model from one device, hospital, skin tone, rhythm mix, or disease prevalence to another.

An association can be useful for prediction even when it is not causal. A causal factor can be difficult to measure directly. A device can be technically accurate yet clinically unhelpful. The packet keeps these axes separate: causal validity, measurement validity, predictive validity, and clinical utility.

Part V

Measurement Science and the Cardiovascular Device Landscape

From Physiological Question to Valid Measurement

A measurement system begins with a question, not a sensor. “Heart health” must be decomposed into explicit measurands and decisions.

26.1 The measurement chain

A rigorous chain has six elements:

1. **Clinical or biological construct:** rhythm, ischemia, cardiac output, arterial stiffness, congestion, plaque burden, or risk.
2. **Physical measurand:** voltage, pressure, optical absorption, acceleration, sound pressure, velocity, concentration, or image intensity.
3. **Sensor and interface:** electrode, photodiode, transducer, microphone, ultrasound array, assay, or catheter.
4. **Signal conditioning and acquisition:** gain, filtering, sampling, quantization, timing, calibration, and protection.
5. **Estimator:** beat detection, interval measurement, waveform model, reconstruction, classifier, or regression.
6. **Interpretation and action:** notification, diagnosis, prognosis, therapy, or reassurance.

Errors can enter at every layer. A model can have excellent accuracy against a biased reference. A highly accurate sensor can estimate the wrong construct. A clinically valid output can still cause harm if delivered without context.

26.2 Accuracy, precision, agreement, and repeatability

- **Accuracy** describes closeness to a reference or true value.
- **Precision** describes spread under repeated conditions.
- **Repeatability** is precision under the same operator, device, site, and short interval.
- **Reproducibility** spans changed operators, devices, sites, or time.
- **Bias** is systematic difference.
- **Limits of agreement** describe the distribution of pairwise differences.
- **Calibration** maps device output to reference scale.
- **Discrimination** ranks or separates states; it does not guarantee calibration.

Correlation is not agreement. Two devices can correlate strongly across a wide physiological range while differing by a clinically unacceptable constant or proportional bias. Bland-Altman analysis is foundational for continuous measurement comparison, but repeated measures and heteroscedastic errors require extensions or mixed-effects approaches (75).

26.3 Reference standards are question-specific

There is no universal reference for “cardiovascular health.” Examples:

Claim	Appropriate reference
Heart rate	Expert-reviewed ECG or validated pulse reference during the tested conditions
AF detection	Interpretable ECG reviewed under a prespecified adjudication process

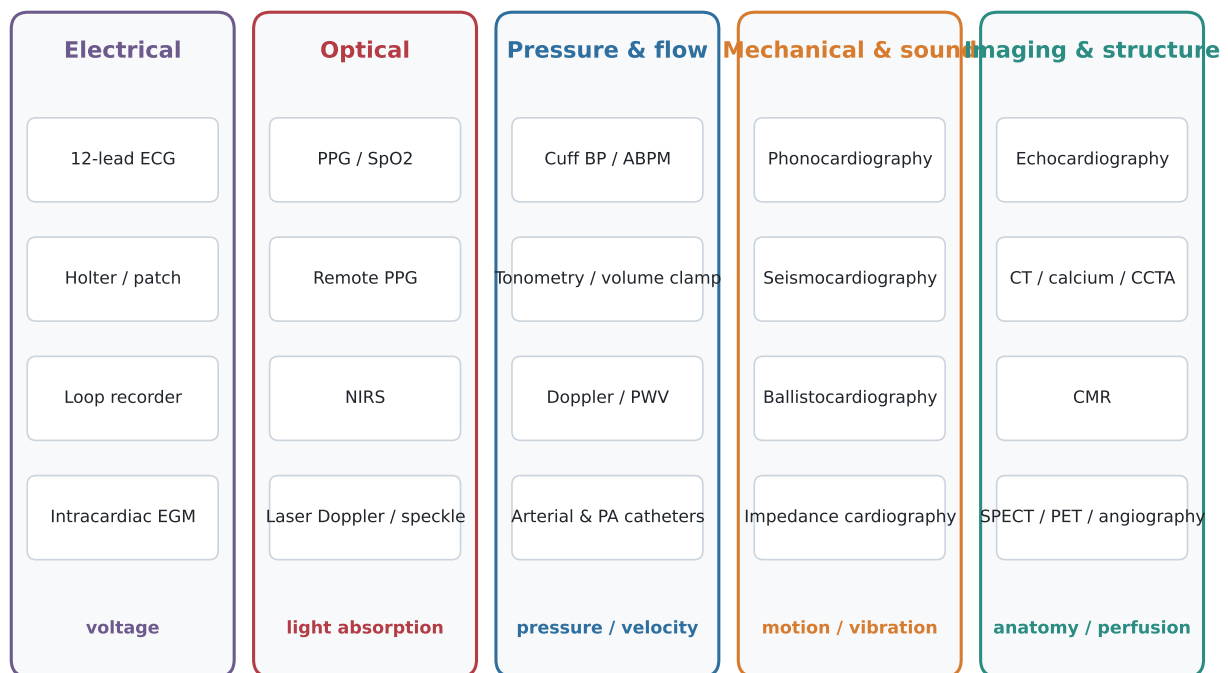
Claim	Appropriate reference
Brachial cuff BP	Validated reference auscultation or invasive reference under applicable protocol
Continuous pressure waveform	Properly leveled, zeroed, and dynamically adequate invasive arterial pressure where ethical/appropriate
SpO2	Arterial co-oximetry across controlled saturation ranges for validation
Ejection fraction	Core-lab echocardiography or CMR under a defined protocol; not interchangeable without analysis
Cardiac output	Direct Fick, thermodilution, or another accepted comparator appropriate to population and setting
Coronary stenosis	Anatomic imaging; physiological significance requires FFR/iFR or validated flow assessment
Sleep apnea	Polysomnography or validated home sleep apnea testing, depending claim

26.4 Sources of variability

Biological variability and measurement error must be separated. Pressure varies beat-to-beat and day-to-day. LDL-C varies with fasting state, illness, and assay. Ejection fraction varies with loading, image quality, and reader. PPG changes with temperature and contact. ECG morphology changes with electrode placement. A valid device report should state:

- within-session quality;
- uncertainty or confidence;
- reference range and population;
- calibration date and method;
- environmental and postural context;
- missingness and exclusion rules;
- firmware, hardware, and algorithm version.

Cardiovascular measurement landscape



Every device is an estimator: define the measurand, reference method, operating envelope, and failure modes.

Figure 26.1: Landscape of cardiovascular measurement modalities organized by physical measurand and degree of invasiveness.

Clinical History, Examination, and Laboratory Devices

27.1 Standardized blood-pressure station

A high-quality BP station includes a validated upper-arm device, multiple cuff sizes, a chair with back support, a table maintaining mid-arm heart level, timed rest, and software that averages repeated readings. The station should record arm, cuff, posture, timing, recent activity, and whether the value is office, unattended, home, or ambulatory (11,34).

27.2 Stethoscope and electronic auscultation

A conventional stethoscope transmits pressure variations through chestpiece, tubing, and earpieces. An electronic stethoscope uses a microphone, piezoelectric element, accelerometer, or pressure sensor, then amplifies and filters the signal. Frequency bands are application-dependent: low-frequency heart sounds and higher-frequency murmurs require different response. Aggressive noise cancellation can remove clinically meaningful transients. Raw, unprocessed audio should be retained for research.

Electronic auscultation can support telemedicine, phonocardiographic visualization, and machine learning. Validation must include varied chest anatomy, sensor pressure, clothing, ambient noise, arrhythmia, valve lesions, and recording sites. A model trained on curated, quiet clips may fail in free living.

27.3 Laboratory instruments

Clinical chemistry analyzers measure lipids, glucose, creatinine, electrolytes, enzymes, and immunoassays. High-sensitivity troponin and natriuretic peptide assays have assay-specific 99th percentiles, analytical ranges, and biological variation. Point-of-care systems trade laboratory throughput and controlled calibration for speed and portability.

Near-patient or wearable biomarker devices are attractive but difficult because interstitial fluid, sweat, saliva, and tears do not automatically track plasma concentration with a fixed lag or ratio. Sweat sodium is useful for cystic fibrosis and hydration research under controlled protocols but is not a direct measure of dietary sodium or blood pressure. Continuous glucose monitoring is the most mature minimally invasive biochemical platform; even there, interstitial lag, compression artifacts, calibration, medications, and sensor drift matter.

27.4 Biomarker interpretation matrix

Biomarker	Primary information	Major limitations
LDL-C	Cholesterol mass in LDL	Particle-number discordance; calculated value limitations
Non-HDL-C	Cholesterol in all apoB particles	Still mass, not particle count
ApoB	Atherogenic particle number proxy	Assay access; treatment thresholds evolving
Lp(a)	Genetically influenced atherogenic particle burden	Assay units/isoform effects; targeted therapies evolving
HbA1c	Medium-term glycemic exposure	Red-cell and hemoglobin effects

Biomarker	Primary information	Major limitations
Creatinine/eGFR	Filtration estimate	Muscle mass, diet, drugs, non-steady state
Urine albumin-creatinine	Glomerular/vascular injury	Exercise, infection, timing; repeat confirmation
hs-troponin	Myocardial injury	Not mechanism-specific; assay/sex/context dependent
BNP/NT-proBNP	Myocardial wall-stress response	Age, AF, kidney disease, obesity, assay effects
hsCRP	Systemic inflammatory risk marker	Nonspecific; acute illness; not a treatment target by itself

Electrocardiographic Devices

28.1 Resting 12-lead ECG

The resting 12-lead ECG remains the reference surface electrical examination. Ten electrodes generate twelve leads. Standard placement is essential: limb-electrode relocation, precordial misplacement, reversed cables, poor contact, or torso placement can change axis, Q waves, ST segments, and rhythm interpretation.

A diagnostic electrocardiograph must preserve morphology, meet amplitude and timing performance, identify lead faults, and withstand expected electrical conditions. IEC 60601-2-25 addresses diagnostic electrocardiographs; monitoring and ambulatory devices have different particular standards (76–78).

28.2 Holter monitors

Traditional Holter systems record continuously, often for 24 to 48 hours, with two or more channels. Modern systems extend duration, storage, and automated analysis. Continuous recording is useful for burden and morphology but generates large data volumes and electrode-maintenance challenges.

Key performance variables include:

- channel count and lead vector;
- sample rate, resolution, and analog bandwidth;
- usable dynamic range and recovery from electrode transients;
- lead-off detection;
- clock accuracy and dropped-sample handling;
- pacemaker pulse detection if claimed;
- validated arrhythmia algorithm performance under AAMI EC57-style testing (15,79).

28.3 Patch monitors

Adhesive patches integrate electrodes, analog front end, battery, memory, and wireless transfer. Advantages are comfort and longer wear; disadvantages include one or two fixed vectors, skin irritation, contact drift, sweat, motion artifact, and inability to reposition electrodes. Diagnostic yield rises with duration for intermittent events, but more duration also increases incidental findings and review workload.

28.4 Event monitors and mobile telemetry

Patient-triggered event recorders store data around symptoms. Auto-triggered loop recorders detect programmed rhythms. Mobile cardiac outpatient telemetry sends events continuously or near-real time. The care pathway matters: an alert has value only if latency, triage, confirmation, and responsibility are defined.

28.5 Implantable loop recorders

Subcutaneous loop recorders monitor for years, enabling detection of rare syncope-related arrhythmia, cryptogenic-stroke AF, or long-term burden. They have a limited sensing vector and are vulnerable to undersensing, oversensing, myopotentials, and noise. Implantation and follow-up make them clinical devices, not DIY platforms.

28.6 Consumer ECG

Smartwatches and handheld devices commonly record a single lead, often analogous to lead I when opposite limbs contact electrodes. Some handheld systems use multiple contact combinations to approximate six limb leads. They can document AF, rate, pauses, ectopy, or symptoms when signal quality is adequate. They cannot reliably exclude ischemia or structural disease, and a normal one-lead recording does not equal a normal 12-lead ECG.

Population studies support scalable AF notification, but performance metrics depend on notification thresholds, wear time, age, disease prevalence, confirmatory testing, and the distinction between algorithm output and adjudicated rhythm (16,17).

28.7 Body-surface mapping and ECG imaging

High-density electrode vests capture many torso potentials. When combined with CT-derived geometry and inverse modeling, electrocardiographic imaging estimates epicardial activation. The inverse problem is ill-posed and depends on regularization, geometry, conductivity assumptions, electrode position, and noise. It is valuable for research and selected electrophysiology applications but is not a direct camera of electrical activation.

28.8 Intracardiac electrograms and EP systems

Catheters record local bipolar and unipolar electrograms, pace tissue, and map activation and voltage. Electroanatomic mapping fuses catheter location, electrograms, and anatomy. Signals are high bandwidth and can contain far-field components. Contact force, orientation, tissue state, filter settings, and annotation alter maps. Ablation systems add therapeutic energy and require extensive safety systems; they are outside any nonclinical build scope.

Engineering a Safe ECG Acquisition Chain

29.1 Electrodes and skin interface

Wet Ag/AgCl electrodes use electrolyte gel to reduce interface impedance and stabilize half-cell potential. Dry metal, conductive polymer, textile, capacitive, and microneedle interfaces improve reusability or comfort but often have higher and more variable impedance, motion sensitivity, polarization, or biocompatibility challenges.

Skin preparation, hair, pressure, sweat, and wear duration change contact. Electrode impedance mismatch converts common-mode interference into differential artifact. A high input impedance helps but cannot fix poor mechanical coupling.

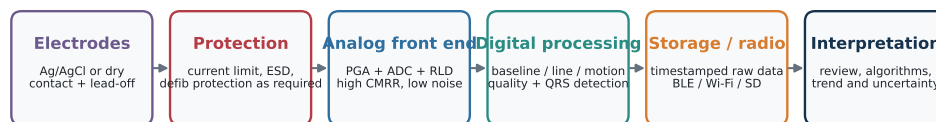
29.2 Analog front end

A representative ECG path contains:

1. input current limiting and electrostatic protection;
2. radiofrequency filtering placed symmetrically;
3. high-input-impedance, low-noise instrumentation or integrated biopotential AFE;
4. common-mode feedback such as right-leg drive with current limiting;
5. anti-alias filtering;
6. high-resolution ADC;
7. galvanically safe data/power architecture;
8. lead-off and saturation monitoring.

Battery-powered ECG acquisition chain

Typical diagnostic ECG bandwidth is broader than fitness heart-rate monitoring; preserve raw data before aggressive filtering.



Safety rail: battery-only human contact during prototyping; no simultaneous mains/USB connection; isolation, leakage, EMC, and risk controls for any clinical intent.

Figure 29.1: Representative ECG analog-front-end chain. Protection, bias return, electrode interface, common-mode management, anti-aliasing, and isolation are all part of the measurement.

Integrated examples include the TI ADS1298 for multichannel biopotential acquisition, Analog Devices MAX30001 for ECG plus bioimpedance, and combined ECG/PPG AFEs such as MAX86176 or TI AFE4950. These are component examples, not complete medical devices; current datasheets, errata, and reference designs must be checked at design freeze (80–83).

29.3 Common-mode and right-leg drive

Mains electric fields couple capacitively to the body and leads. A right-leg-drive amplifier estimates common mode and feeds an inverted, current-limited signal back to the body, reducing common-mode voltage and keeping inputs in range. Stability depends on cable capacitance, electrode impedance, protection networks, and compensation. The circuit must remain safe under single fault.

29.4 Sampling and filtering

For rhythm and morphology research, a practical raw sample rate is often 500 to 1000 samples/s per channel with at least 16 effective bits, though standards and claims determine requirements. Lower rates may be sufficient for rate/rhythm; higher rates help pacemaker pulse detection and detailed QRS research. Store raw or minimally processed data and exact filter metadata.

Common artifact methods include:

- baseline estimation with high-pass, spline, or wavelet methods;
- adaptive mains cancellation;
- motion-quality classification using electrode impedance and IMU;
- robust QRS detection with refractory logic and multi-lead voting;
- ectopy-aware beat alignment;
- signal-quality indices that can suppress interpretation without deleting raw data.

Notch filters can ring and distort QRS/ST morphology. A fixed 50/60 Hz filter does not remove harmonics or electrode-motion artifact. Adaptive filtering requires a reference or model and can remove real signal when assumptions fail.

29.5 Protection and isolation

For any human contact:

- use battery power during attachment;
- disconnect or medically isolate USB, chargers, oscilloscopes, and programmers;
- include current-limiting resistors at every patient lead;
- design for ESD and foreseeable defibrillator exposure only if the claim and standard require it;
- provide creepage/clearance and isolation components appropriate to IEC 60601-1 if crossing a patient isolation boundary;
- test leakage and fault conditions with proper equipment;
- begin with ECG simulators, not people.

A battery-powered front end becomes mains-referenced as soon as an unisolated USB, oscilloscope ground, bench supply, programmer, or charger is connected. During human recording, use an independently battery-powered logger or a properly designed medical isolation barrier. "Only 5 V" does not establish patient safety.

Blood Pressure: Cuffs, Continuous Systems, Tonometry, and Cuffless Estimation

30.1 Auscultatory measurement

A cuff occludes the brachial artery. During controlled deflation, Korotkoff sounds appear and change as flow transitions through the partially compressed artery. The first sound is used for systolic pressure and disappearance for diastolic pressure under standard conditions. Observer hearing, deflation rate, stethoscope placement, arrhythmia, movement, cuff size, and terminal digit preference contribute error.

30.2 Oscillometric measurement

Automated cuffs detect small pressure oscillations superimposed on cuff pressure. Oscillation amplitude typically increases, reaches a maximum near mean arterial pressure, then decreases. Proprietary algorithms estimate systolic and diastolic pressure from the envelope shape, ratios, derivatives, and other features. The ratios are not universal physical constants.

Oscillometric blood-pressure measurement

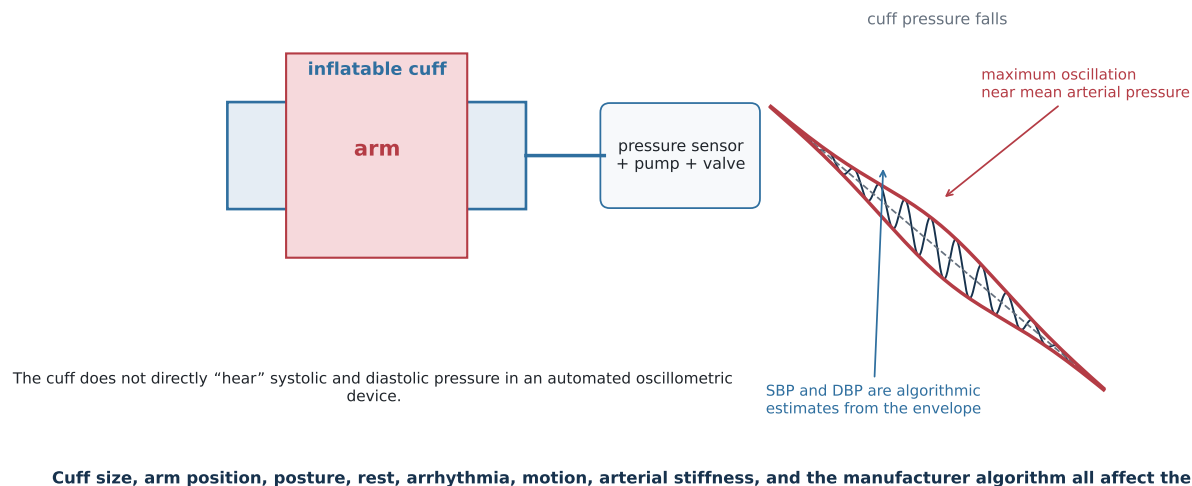


Figure 30.1: Oscillometric cuff principle: cuff pressure, arterial oscillation envelope, and algorithmic estimation of systolic, mean, and diastolic pressure.

Engineering elements include:

- cuff and bladder geometry;
- pump, dump valve, proportional valve or bleed orifice;
- pressure sensor covering the required range and overpressure margin;
- pneumatic leak and occlusion detection;
- low-noise sampling of static pressure plus small pulsations;
- controlled inflation/deflation trajectory;
- arrhythmia and motion handling;

- safety cutoff and mechanical pressure relief;
- algorithm validation across arm size, age, pressure, rhythm, pregnancy, and disease.

A home prototype can explore oscillometry, but a clinically usable device must satisfy applicable performance and clinical-validation standards such as IEC 80601-2-30 and ISO 81060-2 (84,85).

30.3 Ambulatory blood pressure monitoring

ABPM inflates periodically over 24 hours or longer. It provides awake, asleep, morning, and variability information. Errors arise from motion, failed readings, incorrect sleep windows, cuff discomfort, and behavior change. A valid report shows the number and proportion of successful readings, time windows, averages, and quality criteria.

30.4 Finger volume-clamp systems

The Penaz/vascular-unloading method uses a finger cuff, optical PPG, and fast servo loop to maintain near-constant arterial volume. The cuff pressure required to unload the artery estimates a continuous finger pressure waveform. Calibration and transfer to brachial pressure are necessary; vasoconstriction, temperature, finger size, movement, hydrostatic height, and vascular disease can degrade performance.

30.5 Applanation tonometry

A pressure sensor gently flattens a superficial artery such as radial or carotid, reducing circumferential wall force so surface pressure approximates intra-arterial pressure shape. Calibration usually uses cuff pressure. Operator position and hold-down force matter. Transfer functions may estimate central aortic waveforms, but this adds model uncertainty.

30.6 Cuffless blood pressure

Cuffless systems use PPG morphology, ECG-PPG timing, pulse-wave velocity, tonometry, radar, ultrasound, or combinations. The most common timing relation is

$$PAT = PEP + PTT,$$

where pulse-arrival time from ECG to peripheral pulse includes pre-ejection period plus pulse-transit time. Because PEP changes with sympathetic state, contractility, preload, and exercise, ECG-to-PPG timing is not pure vascular transit.

The relation between BP and transit time is individual, nonlinear, pressure-state dependent, and altered by temperature, vasoactive state, age, disease, posture, and sensor site. Calibration can hide these dependencies for a short window but drift when physiology changes. This is why current professional statements do not recommend unvalidated cuffless devices for diagnosis or treatment, and why the FDA issued draft performance and labeling guidance in January 2026 (86,87).

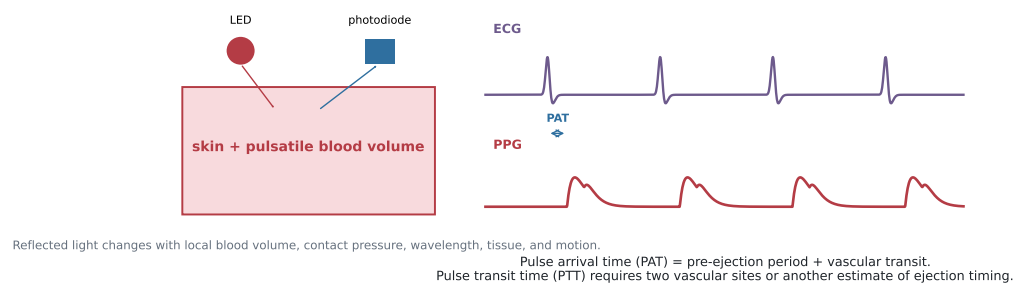
30.7 Hydrostatic effects

A sensor height difference h from the heart creates approximately

$$\Delta P = \rho gh.$$

For blood, this is roughly 0.77 mmHg per centimeter of vertical height. A wrist 20 cm below the heart may read about 15 mmHg higher solely from hydrostatic pressure. Any wrist pressure or PPG-derived system needs reliable position estimation or explicit user positioning.

PPG, pulse arrival, and the cuffless-BP temptation



BP inference is person-, posture-, temperature-, tone-, and calibration-dependent; synchronized ECG+PPG does not make it reference-grade by itself.

Figure 30.2: ECG-to-PPG timing, pulse arrival, pre-ejection period, pulse-transit time, and calibration dependencies.

Photoplethysmography and Pulse Oximetry

31.1 PPG physics

Photoplethysmography illuminates tissue and detects transmitted or reflected light. The signal contains a large DC component from tissue, venous blood, nonpulsatile arterial blood, and geometry, plus a smaller AC component related mainly to pulsatile arterial volume and optical path changes. The Beer-Lambert law provides intuition, but scattering, heterogeneous tissue, contact pressure, and geometry make tissue PPG more complex than a uniform absorbing solution (88).

Common wavelengths include green for strong superficial pulsatile signal at the wrist, red and infrared for oxygen saturation, and sometimes additional wavelengths for deeper or multi-constituent analysis. LED-photodiode spacing influences penetration and signal strength.

31.2 Reflective versus transmissive configurations

Transmissive PPG places emitter and detector across tissue, as in a fingertip or earlobe. Reflective PPG places them on the same surface, enabling wrist and patch devices. Transmissive geometry often provides a larger pulsatile component but is limited to suitable sites. Reflective PPG is more sensitive to placement, pressure, motion, and ambient light.

31.3 Pulse oximetry

Pulse oximetry estimates arterial oxygen saturation from the differential pulsatile absorption of red and infrared light. A ratio of ratios is mapped through empirical calibration:

$$R = \frac{AC_{red}/DC_{red}}{AC_{IR}/DC_{IR}}.$$

The mapping depends on device optics, calibration data, dyshemoglobins, perfusion, motion, and tissue. Standard two-wavelength pulse oximeters cannot separately identify carboxyhemoglobin and methemoglobin. Accuracy worsens at low saturation, low perfusion, motion, venous pulsation, nail coatings, poor contact, and some skin-pigmentation conditions.

The discovery of clinically meaningful occult hypoxemia disparities in darker-skinned patients demonstrated why calibration and subgroup validation are not optional (72). Pulse-oximeter validation is governed by standards such as ISO 80601-2-61 and regulatory guidance (89).

31.4 PPG-derived features

PPG can support:

- pulse rate and rhythm irregularity;
- pulse amplitude and perfusion trends;
- respiratory modulation;
- pulse upstroke and contour features;
- intersite transit timing;
- pulse-rate variability under clean conditions;
- estimates of arterial stiffness or pressure in calibrated models;
- vascular occlusion and reactive-hyperemia studies.

Many morphology features are sensitive to sampling rate, analog filtering, LED current, contact pressure, temperature, and site. Feature definitions should be explicit and measured on raw or standardized waveforms.

31.5 Motion and ambient rejection

A high-quality optical design uses:

- alternating LED phases plus dark/ambient samples;
- transimpedance amplifier dynamic range for large DC and small AC;
- optical barriers to reduce direct LED-to-photodiode crosstalk;
- mechanical control of contact pressure and stray light;
- synchronized IMU data;
- saturation and clipping detection;
- per-wavelength quality indices;
- adaptive LED current and receiver gain with logged state.

Adaptive motion cancellation is difficult because motion changes both additive artifact and the tissue optical transfer function. An accelerometer reference is correlated but not identical. It is safer to withhold interpretation during poor quality than to output a confident fabricated pulse.

Remote PPG, Cameras, Radar, and Contactless Monitoring

32.1 Remote photoplethysmography

Remote PPG estimates subtle color changes in video, commonly from the face. Algorithms combine color channels, spatial regions, skin segmentation, illumination normalization, and motion compensation. It can estimate pulse under controlled light and low motion. Performance degrades with head movement, compression, frame-rate instability, autoexposure, rolling shutter, low light, makeup, skin tone imbalance, and changing display illumination.

Contactless respiratory rate and pulse can be useful for trends or telehealth, but clinical claims require device-specific validation. Video storage also creates biometric and privacy risks beyond a simple heart-rate value.

32.2 Radar and radiofrequency sensing

Doppler radar and frequency-modulated continuous-wave radar can detect chest-wall displacement from respiration and heartbeat. Multiple-input/multiple-output arrays can localize subjects. Clothing, posture, multipath, cohabitants, body movement, and reflections complicate extraction. Radar measures displacement or phase change, not an ECG, and cannot assign every beat reliably under motion without sophisticated quality control.

32.3 Thermal imaging

Infrared cameras estimate surface temperature and can track breathing around the nose or perfusion changes. Surface temperature depends on environment, emissivity, blood flow, and distance. It is not core temperature or direct cardiac output.

Mechanical and Acoustic Cardiography

33.1 Phonocardiography

Phonocardiography digitizes heart sounds and murmurs. Useful features include S1/S2 timing, splitting, additional sounds, spectral energy, and murmur timing. ECG synchronization helps define systole and diastole. Microphone response, chestpiece geometry, pressure, placement, body habitus, lung sounds, and ambient noise must be controlled.

33.2 Seismocardiography

Seismocardiography records chest-wall acceleration produced by cardiac mechanical events. Tri-axial accelerometers capture complex motion; gyroscopes measure rotational components. Fiducial points have been associated with valve opening/closure, ejection, filling, and contractility, but labels vary by sensor location, posture, respiration, and disease. The signal is not a direct valve-motion trace.

33.3 Ballistocardiography

Ballistocardiography measures whole-body recoil from blood acceleration, using beds, chairs, scales, or suspended platforms. It is attractive for unobtrusive home monitoring but sensitive to posture, mattress mechanics, body movement, and transfer function. It can estimate beat timing and trends; absolute stroke volume is difficult without calibration and modeling.

33.4 Gyrocardiography and wearable ultrasonics

Gyrocardiography measures angular chest motion and can complement accelerometers. Flexible ultrasound patches can image tissue motion or flow and may eventually bridge wearables and echocardiography, but coupling, acoustic power, beam steering, heating, motion, and interpretation remain engineering challenges.

33.5 Electromechanical intervals

Combining ECG with SCG, PCG, impedance, or PPG yields intervals such as:

- QRS onset to aortic-opening surrogate;
- pre-ejection period;
- left-ventricular ejection time;
- systolic time ratio;
- pulse arrival at peripheral sites;
- respiratory modulation.

These can track loading or contractility under controlled conditions but are not universally diagnostic. Fiducial uncertainty should be propagated rather than ignored.

Bioimpedance and Impedance Cardiography

34.1 Basic principle

A small alternating current is injected through the body, and resulting voltage estimates complex impedance. Tissue fluids conduct; membranes add capacitive behavior. Electrode geometry and frequency determine sampled volume. Four-electrode configurations reduce contact-impedance influence compared with two-electrode methods.

34.2 Thoracic impedance and cardiac output

Impedance cardiography tracks beat-related changes in thoracic impedance, often attributed to aortic blood-volume and velocity changes. Models estimate stroke volume using body geometry, baseline impedance, ejection time, and the maximum derivative dZ/dt . Different formulas make assumptions about thoracic shape and current distribution.

Errors arise from obesity, edema, lung fluid, posture, electrode placement, valve disease, arrhythmia, aortic pathology, ventilation, and motion. Trending within a person may be more robust than absolute cardiac output. Bioreactance analyzes phase shifts and has similar validation needs.

34.3 Fluid-status monitoring

Thoracic impedance often falls when conductive fluid increases, but lung fluid is not the only determinant. Implantable devices have used intrathoracic impedance as one input for heart-failure alerts; specificity is limited. Multi-frequency spectroscopy can distinguish extracellular and intracellular components in limbs or whole body, but translating this to pulmonary congestion requires careful validation.

Vascular Function, Flow, and Perfusion Devices

35.1 Ankle-brachial index

ABI is the ratio of ankle systolic pressure to brachial systolic pressure. It can be measured with Doppler and cuffs or selected automated methods. A low ABI indicates lower-extremity arterial obstruction and systemic risk. A high or noncompressible ABI can reflect medial calcification; toe-brachial index may be more useful (48,49).

Protocol details matter: rest, cuff width, Doppler angle, artery selection, sequence, and whether the higher or lower ankle artery is used for the clinical definition. Exercise ABI can reveal flow limitation when resting ABI is normal.

35.2 Pulse-wave velocity

Carotid-femoral PWV is a reference noninvasive measure of aortic stiffness. It divides an estimated path length by transit time between waveforms. Distance measurement is a major source of systematic variation; waveform foot algorithms and gating also matter (21).

A research device can use tonometry, pressure cuffs, PPG, or Doppler at two sites. It must disclose path-length method and whether simultaneous or sequential acquisition was used. Blood pressure at the time of PWV is essential because stiffness is pressure-dependent.

35.3 Flow-mediated dilation

Brachial FMD uses ultrasound to measure arterial dilation after a period of forearm occlusion. The response depends on shear stimulus, baseline diameter, cuff location, image quality, edge detection, timing, temperature, fasting, medications, and recent activity. Standardized methodology and shear normalization are necessary (20).

35.4 Laser Doppler and speckle

Laser Doppler flowmetry measures frequency shifts from moving red cells in a small tissue volume, yielding relative perfusion units. Laser speckle contrast imaging maps flow over a surface. Both are sensitive to motion, temperature, probe pressure, skin properties, and superficial circulation. They are valuable for microvascular research but do not directly measure central cardiovascular health.

35.5 Venous and jugular signals

Jugular venous pressure and waveform reflect right-atrial pressure and events, modified by respiration, posture, valve disease, rhythm, and neck anatomy. Ultrasound, camera, PPG, or pressure sensors can estimate jugular pulsation. Separating jugular from carotid signals and establishing absolute height remain difficult. A trend may be useful, but a black-box “fluid score” needs reference validation.

Echocardiography and Ultrasound

36.1 Two-dimensional and M-mode imaging

Echocardiography transmits ultrasound pulses and forms images from returning echoes. Axial resolution depends on pulse length; lateral resolution depends on beam width; frame rate trades against depth, sector width, and line density. M-mode samples one line at high temporal resolution.

Two-dimensional echo evaluates chamber size, wall thickness, global and regional motion, valves, pericardium, masses, and great vessels. Image quality depends on acoustic windows, body habitus, lung, probe frequency, and operator skill.

36.2 Doppler

Pulsed-wave Doppler measures velocity at a selected sample volume but aliases above the Nyquist limit. Continuous-wave Doppler measures high velocities along the entire beam but lacks range localization. Color Doppler maps mean velocity and variance estimates. The Doppler equation is

$$\Delta f = \frac{2f_0 v \cos \theta}{c},$$

so angle error grows rapidly when the beam is not aligned with flow. The simplified Bernoulli relation estimates pressure gradients from peak velocity.

36.3 Tissue Doppler and strain

Tissue Doppler measures myocardial velocities but is angle-dependent. Speckle-tracking strain follows natural acoustic patterns to estimate deformation. Global longitudinal strain can detect dysfunction before EF changes in some diseases, but vendor, software, image quality, loading, and tracking quality affect values.

36.4 Chamber quantification

Volumes are derived from geometric models or three-dimensional datasets. Foreshortening underestimates left-ventricular volume. EF reproducibility is limited by border definition and loading. CMR may be more reproducible for volumes in selected patients, but echo is more portable and provides hemodynamics (90).

36.5 Point-of-care ultrasound

Handheld ultrasound can extend examination for ventricular function, congestion, pericardial effusion, and aortic or venous questions. Training and scope are critical. AI guidance can improve image acquisition, but automated labels should not exceed the quality of the acquired view.

Cardiac CT, Coronary Calcium, and CT Angiography

37.1 Coronary artery calcium

Noncontrast ECG-gated CT quantifies calcified coronary plaque, commonly using the Agatston score. Calcium burden strongly predicts atherosclerotic risk and can reclassify preventive decisions. It does not measure noncalcified plaque or stenosis and exposes the patient to ionizing radiation, though contemporary doses are low.

A score of zero indicates low near-term event risk in many asymptomatic populations but is less reassuring in young adults, smokers, diabetes, severe hypercholesterolemia, strong family history, or symptoms. Serial scanning is not automatically useful because treatment can alter plaque density and progression interpretation.

37.2 Coronary CT angiography

CCTA uses iodinated contrast, rapid gantry rotation, ECG gating, and reconstruction to visualize lumen and wall plaque. It can identify stenosis, noncalcified plaque, remodeling, and high-risk features. Heart rate, rhythm, calcification, obesity, motion, scanner capabilities, and contrast timing affect quality.

CAD-RADS standardizes reporting and adds plaque and ischemia modifiers (91). CT-derived FFR applies computational or machine-learning models to CCTA anatomy to estimate physiology; it is an inference, not a direct pressure measurement.

37.3 Radiation and contrast

Dose depends on protocol, patient size, scanner, gating, and scan length. Iodinated contrast can cause allergic-like reactions and kidney risk in susceptible contexts. Devices and algorithms should report dose metrics and quality rather than only the final image.

Cardiovascular Magnetic Resonance

CMR uses magnetic fields and radiofrequency excitation to characterize anatomy, function, flow, perfusion, and tissue without ionizing radiation. Cine imaging quantifies volumes and motion. Phase-contrast imaging measures velocity and flow. T1/T2 mapping, late gadolinium enhancement, and extracellular-volume estimates characterize fibrosis, edema, infiltration, and scar.

CMR is particularly valuable for cardiomyopathy, myocarditis, viability, congenital disease, masses, pericardium, and right-ventricular assessment. Limitations include cost, availability, scan duration, arrhythmia or breath-hold challenges, implants, claustrophobia, and contrast considerations. Standardized protocols improve reproducibility (92).

CMR-derived tissue signatures are not pathognomonic. Late enhancement patterns narrow differential diagnosis but must be integrated with clinical data. Parametric maps are sequence-, field-, scanner-, and site-dependent; local reference ranges and quality control matter.

Nuclear Imaging, PET, and Molecular Measurement

SPECT myocardial perfusion imaging uses gamma-emitting tracers and a gamma camera to assess relative perfusion under stress and rest. PET uses positron-emitting tracers and can quantify absolute myocardial blood flow and coronary flow reserve with appropriate protocols. PET also measures metabolism, inflammation, amyloid, and viability using specific tracers.

Relative perfusion can miss balanced multivessel reduction because the best-perfused region becomes the reference. Attenuation, motion, extracardiac activity, obesity, and reconstruction influence images. Absolute flow adds physiological information but requires tracer-specific kinetic modeling and quality control.

Nuclear imaging involves ionizing radiation and radiopharmaceutical logistics. It provides biology not available from surface wearables; the modalities are complementary.

Catheter Angiography, Intravascular Imaging, and Invasive Hemodynamics

40.1 Coronary angiography

Catheter angiography injects contrast directly into coronary arteries and images the lumen with x-rays. It has high temporal and spatial resolution and allows immediate intervention. It shows a luminogram rather than the full plaque wall and can underestimate diffuse disease.

40.2 IVUS and OCT

Intravascular ultrasound images vessel wall and plaque with deeper penetration and lower axial resolution than optical coherence tomography. OCT uses near-infrared light for high-resolution near-surface imaging but requires blood displacement and has shallower penetration. They quantify stent expansion, apposition, plaque morphology, and vessel dimensions.

40.3 Pressure wires and coronary physiology

FFR, iFR, coronary flow reserve, and microvascular resistance interrogate different components. FFR assesses pressure loss under hyperemia; iFR uses resting diastole; flow reserve combines epicardial and microvascular effects; resistance indices focus more on microcirculation. Values should not be treated as interchangeable.

40.4 Right-heart catheterization

A right-heart study measures right atrial, right ventricular, pulmonary artery, and pulmonary capillary wedge pressures; oxygen saturations; and cardiac output. Waveform interpretation, end-expiratory measurement, leveling, zeroing, catheter position, and respiratory mechanics are critical. Wedge pressure is an estimate of left-atrial pressure only when a valid occluded pulmonary arterial segment is obtained without artifact.

40.5 Arterial lines

A fluid-filled catheter-transducer system has natural frequency and damping. Under-damping exaggerates systolic peaks and ringing; over-damping blunts systolic pressure and raises diastolic pressure, while mean pressure may be less affected.

The fast-flush square-wave test evaluates response. Tubing length, compliance, air bubbles, clots, stopcocks, and catheter whip alter dynamics. The transducer must be leveled to the reference point and zeroed to atmosphere. IEC 60601-2-34 addresses invasive pressure monitoring equipment (93).

40.6 Pressure-volume catheters

Conductance catheters estimate ventricular volume from intracavitary electrical conductance while measuring pressure. Parallel conductance, field inhomogeneity, geometry, and calibration create error. They are research and advanced clinical tools requiring invasive expertise, not a build target for a general engineering lab.

Fluid-filled arterial-line dynamics: same artery, different displayed waveform

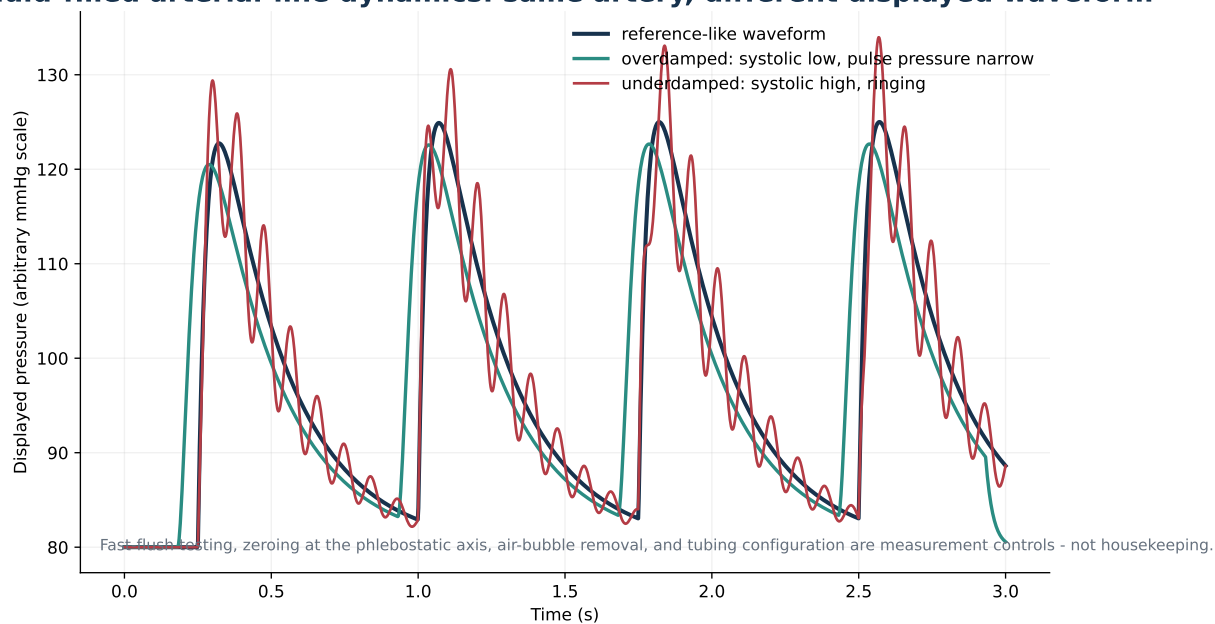


Figure 40.1: Dynamic response of invasive arterial pressure systems: adequate response, under-damping/ringing, and over-damping.

Implantable and Therapeutic Device Telemetry

Pacemakers, ICDs, cardiac resynchronization devices, and implantable monitors record electrograms, rates, arrhythmia episodes, pacing percentage, lead impedance, battery status, and sometimes thoracic impedance or heart-sound surrogates. These devices provide valuable longitudinal data but are optimized for therapy and safety, not unrestricted research access.

Implantable pulmonary artery pressure systems provide daily hemodynamic trends in selected heart-failure patients. Other investigational implants target left-atrial pressure, vascular pressure, or multi-sensor decompensation. Implantable data require secure telemetry, long-term biocompatibility, hermetic packaging, battery or passive power, drift control, and a clinical response program (45).

A noninvasive research project should treat implantable systems as reference landscape, not a DIY extension. Infection, thrombosis, embolization, migration, tissue injury, and device failure make implant development a specialized regulated program.

Exercise Testing and Functional Measurement

42.1 Exercise ECG

Treadmill or bicycle testing records ECG, symptoms, workload, heart rate, and pressure. It can identify exercise-induced ischemia, arrhythmia, chronotropic response, and functional capacity. Baseline ECG abnormalities, medication, pretest probability, and test quality affect interpretation. Imaging stress tests are preferred for selected patients.

42.2 Cardiopulmonary exercise testing

CPET measures ventilation, oxygen uptake, carbon dioxide output, gas exchange, and ECG. Key variables include peak VO_2 , ventilatory thresholds, oxygen pulse, VE/VCO_2 slope, breathing reserve, and recovery. It can differentiate cardiac, pulmonary, circulatory, muscular, and effort limitations.

Wearable VO_2 estimates use pace or power plus heart rate and demographic models. They are useful for trends in a consistent activity but should not be labeled direct oxygen consumption unless validated against metabolic measurement.

42.3 Six-minute walk and field tests

The six-minute walk captures integrated submaximal function but depends on corridor, instructions, learning, motivation, oxygen, aids, and disease. Step tests, sit-to-stand, and walking speed provide accessible functional data. A home system should standardize route, device location, start condition, symptoms, and environment.

Part VI

Engineering Cardiovascular Measurement Systems

Start With the Intended Claim

A device architecture should be derived from the intended use and claim. “Monitor cardiovascular health” is not testable. Better claims are narrow:

- acquire research-quality single-lead ECG during rest and walking;
- estimate pulse rate during daily living within specified error and quality conditions;
- measure upper-arm oscillometric blood pressure under a defined protocol;
- quantify ECG-to-PPG timing during a standardized posture challenge;
- detect episodes of possible irregular rhythm and preserve confirmatory ECG;
- trend electromechanical intervals under controlled exercise and recovery;
- measure bilateral pulse arrival and ankle-brachial pressure in a supervised station.

Each claim determines the reference, population, bandwidth, sample rate, wear duration, acceptable failure rate, and regulatory pathway.

43.1 Design-control traceability

A practical traceability chain is:

User need → System requirement → Subsystem requirement → Verification → Clinical validation.

Example:

Level	Example for ambulatory ECG
User need	A clinician needs interpretable rhythm around intermittent palpitations.
System requirement	Record one ECG vector continuously for 7 days and preserve events with less than a specified data-loss rate.
Hardware requirement	Input noise, dynamic range, battery life, electrode wear, storage, clock, and lead-off detection meet specified limits.
Algorithm requirement	QRS detection and AF candidate logic meet prespecified sensitivity, PPV, episode duration, and quality-gating criteria.
Verification	Simulator, injected waveforms, EMC, mechanical, battery, temperature, and fault tests.
Validation	Prospective paired recording against adjudicated reference ECG in the intended population.

A device is not “90% accurate” without defining which output, denominator, population, prevalence, and confidence interval.

Recommended Multisensor Research Architecture

A flexible first-generation platform should prioritize raw synchronized signals, safety, modularity, and testability over miniaturization.

Reference architecture for a research cardiovascular patch

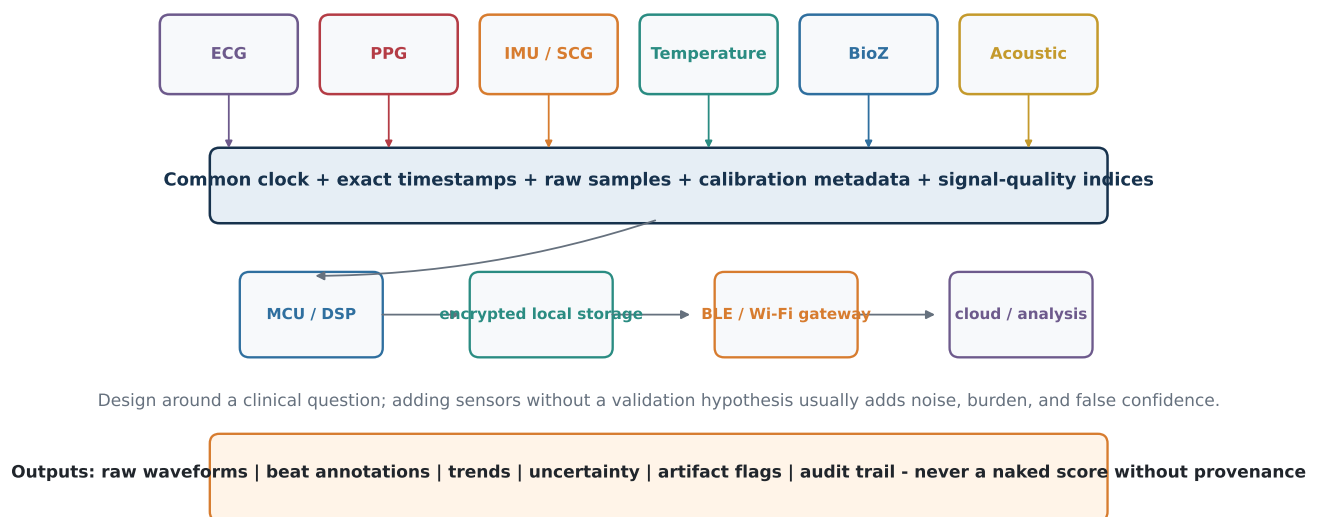


Figure 44.1: Reference architecture for synchronized ECG, PPG, inertial, temperature, acoustic, pressure, and contextual acquisition.

44.1 Core node

A practical wearable or chest-patch node includes:

- one or two ECG channels with lead-off and saturation detection;
- two or more PPG wavelengths, ambient sampling, programmable LED current, and receiver gain;
- low-noise 3-axis accelerometer and gyroscope;
- skin-contact and ambient temperature;
- optional microphone or contact accelerometer for PCG/SCG;
- optional bioimpedance channel;
- monotonic high-resolution hardware timer shared across channels;
- local flash storage with checksums and a power-loss-safe file system;
- low-power radio for metadata, live preview, and transfer;
- battery, fuel gauge, charge protection, and hard power separation from patient contact.

44.2 Station node

A supervised benchtop station can add:

- 6- or 12-lead ECG;
- validated clinical BP monitor plus research cuff pressure tap or independent cuff system;
- digital stethoscope at standardized sites;
- bilateral finger/toe PPG;
- Doppler probe for ankle pressure and flow timing;
- weight scale and optional ballistocardiography;
- spirometry or metabolic cart integration;
- trigger I/O to synchronize ultrasound or external reference devices.

44.3 Environmental and contextual node

Context can include:

- ambient temperature, humidity, light, sound level, and barometric pressure;
- air-quality data from a calibrated local monitor or authoritative geospatial service;
- posture, location class, activity type, medication time, meal, caffeine, alcohol, symptoms, and sleep window;
- device fit, sensor site, and quality flags.

Context data should be privacy-minimized. Store derived environment classes when precise GPS is unnecessary.

44.4 Why raw data matters

Firmware will change. Algorithms will improve. If only a proprietary “stress score” or beat table is stored, future analysis is impossible. Raw data should be retained at least for development cohorts, with separate derived tables for efficient use. Every file should include hardware revision, firmware version, algorithm version, calibration constants, sample-rate measurements, and time-base metadata.

Signal and System Specifications

The values below are reasonable research starting points, not universal regulatory specifications.

Channel	Typical raw sample rate	Useful analog bandwidth	Resolution / notes
Diagnostic-style ECG	500-1000 Hz	approximately 0.05-150 Hz	16-24-bit ADC/AFE; preserve morphology
Rhythm ECG	250-500 Hz	approximately 0.5-40 Hz, plus raw if possible	Narrower filters may be acceptable for rhythm claim
PPG	100-500 Hz	DC to approximately 20 Hz	Higher rate helps timing and morphology; log LED/gain state
SCG/gyrocardiography	500-2000 Hz	approximately 0.5-100 Hz	Low-noise IMU; rigid mechanical coupling
PCG	2-8 kHz	approximately 20-1000+ Hz	Application-specific microphone response
Cuff pressure	100-500 Hz	DC plus pulse oscillations	Sensor range commonly at least 0-300 mmHg with margin
Bioimpedance waveform	100-1000 Hz demodulated	Beat/respiratory band	Carrier and electrode scheme documented
Skin temperature	1-10 Hz	Slow	Thermal settling and ambient compensation
Activity IMU	50-200 Hz	DC to tens of Hz	Higher for impact or SCG; anti-aliasing required
Audio annotation	16-48 kHz	Voice band	Privacy-sensitive; avoid unless justified

45.1 Clock quality

Timing precision is often more important than amplitude precision for multimodal fusion. A 100 ppm clock can drift about 8.6 seconds per day. For ECG-PPG timing in milliseconds, channels need a shared hardware clock or frequent robust synchronization. BLE packet arrival time is not acquisition time.

Recommended approach:

- timestamp samples at acquisition with a hardware timer;
- use one oscillator for all on-node ADCs when possible;
- record nominal and measured sample rate;
- emit periodic synchronization markers;
- for multiple nodes, use wired trigger during lab validation or two-way clock synchronization with drift estimation;
- never interpolate away dropped data without a gap flag.

45.2 Dynamic range and saturation

Movement and electrode transients can exceed physiological amplitude by orders of magnitude. Design for rapid overload recovery, not merely small-signal noise. PPG ambient light and contact changes can saturate the receiver. Automatic gain can preserve signal but changes morphology; every gain or LED-current transition must be logged.

45.3 Noise budgeting

A signal budget should include:

- sensor intrinsic noise;
- electrode or photodiode shot noise;
- amplifier voltage and current noise;
- resistor thermal noise;
- ADC quantization and reference noise;
- power-supply ripple;
- digital coupling;
- motion and interface artifact;
- environmental electromagnetic interference.

Do not specify only ADC bits. Effective resolution is set by the complete chain and usable dynamic range.

PCB, Power, and Mechanical Design

46.1 Partitioning and return paths

Separate low-level analog, high-current LED/pump/valve, radio, memory, and switching-power regions. Partitioning does not mean splitting ground indiscriminately; current return paths should be short and controlled. Avoid routing digital clocks under electrode inputs, photodiode nodes, or transimpedance amplifiers.

Place patient protection and RF input components at the connector. Place decoupling at every supply pin with the recommended values and loop area. Provide test points for rails, reference voltages, bias, electrode common mode, analog outputs, clock, and critical digital buses. Make rails isolatable through zero-ohm links or current-measurement jumpers during bring-up.

46.2 Power rails

Create a power budget for every mode:

- sleep;
- acquisition;
- peak PPG LED current;
- radio advertising and transfer;
- flash write;
- cuff pump and valve operation;
- audio acquisition;
- charging.

Measure, do not infer, quiescent and peak current. Brownout during LED pulse or flash write can corrupt data. Fuel-gauge state of charge should be validated across temperature and cell age.

46.3 Mechanical interface

Signal quality often depends more on mechanics than electronics.

- ECG electrodes need stable skin contact without excessive pressure or irritation.
- PPG requires controlled optical contact, barrier walls, limited ambient leakage, and a pressure range that avoids venous occlusion or tissue blanching.
- SCG requires a rigid, repeatable coupling site; a loose enclosure acts as a mechanical filter.
- PCG chestpiece geometry and pressure shape frequency response.
- Wrist devices need anti-rotation features and validated strap tightness.
- Cuff bladder dimensions and placement must match arm circumference.

Enclosure design should separate sweat paths from high-impedance nodes, allow cleaning, and avoid sharp edges. Adhesives and skin-contact materials require biocompatibility evaluation for the type and duration of contact (94).

46.4 Connectors and debug access

Avoid exposed patient-connected metal that can contact a charger or bench equipment. Use keyed connectors and clear labeling. A development board can include debug headers, but production/human-use configurations should physically prevent unsafe simultaneous connections. Debug and manufacturing access should not depend on soldering to sensitive analog nodes.

ECG Algorithms and Quality Control

47.1 QRS detection

A robust QRS detector uses band-limited slope or energy, adaptive thresholds, refractory periods, and search-back logic. Multi-channel systems can vote across leads. False detections arise from muscle artifact, T waves, pacing pulses, and electrode transients; missed beats arise from low amplitude, broad QRS, saturation, and premature beats.

Evaluation should be beat-level with tolerance windows and event-level for rhythms. Report false positives per hour, not only sensitivity, because long monitoring makes a small false-positive rate burdensome.

47.2 Rhythm classification

Atrial fibrillation algorithms combine irregularity, absence or inconsistency of P waves, morphology, and episode duration. PPG algorithms primarily detect pulse irregularity and cannot distinguish all causes. Ectopy, sinus arrhythmia, motion, tremor, and frequent premature beats can mimic AF.

A safe workflow is:

1. compute signal quality;
2. detect beats with confidence;
3. generate candidate episode;
4. preserve raw window and context;
5. seek ECG confirmation or expert review;
6. report uncertainty and episode burden, not a single binary lifetime diagnosis.

47.3 HRV

Time-domain metrics include SDNN and RMSSD; frequency-domain metrics partition power into bands; nonlinear metrics examine complexity and geometry. Interpretation requires stationary segments, ectopy handling, duration, posture, respiration, and age. PPG-derived pulse-rate variability differs from ECG HRV because pulse transit and missed peripheral pulses add variability.

A consumer “stress” inference from HRV can be directionally useful within one person under controlled conditions, but it is not a validated measure of psychological stress or autonomic health in isolation.

47.4 Ischemia and morphology

ST analysis requires stable electrode placement, low-frequency fidelity, posture awareness, and careful baseline selection. Ambulatory ischemia algorithms are vulnerable to lead motion, rate-related repolarization, bundle-branch block, pacing, hypertrophy, and artifact. A one-lead wearable should not claim to rule out acute coronary syndrome.

PPG, SpO2, and Optical Algorithms

48.1 Pulse detection and quality

Beat detection can use the first derivative, matched filters, adaptive thresholds, or model fitting. Quality indices should include:

- AC/DC ratio;
- saturation and clipping;
- spectral concentration around pulse rate;
- agreement across wavelengths or photodiodes;
- motion coherence;
- template consistency;
- perfusion level;
- implausible interval or morphology transitions.

Output should be absent or explicitly low-confidence when quality fails.

48.2 Oxygen-saturation estimation

SpO2 validation requires controlled hypoxemia studies with arterial co-oximetry under applicable standards and ethical oversight. Bench optical phantoms cannot replace human calibration because scattering and physiology matter. Subgroup analysis should include skin pigmentation across the intended population, low perfusion, motion, and sensor sites (72,89).

48.3 Pulse morphology

Features such as rise time, systolic peak, dicrotic notch, reflection index, stiffness index, and area ratios are not standardized across filters and sites. A feature-development pipeline should:

1. define signal preprocessing;
2. define fiducials mathematically;
3. quantify detection uncertainty;
4. assess repeatability and pressure/temperature dependence;
5. test incremental value beyond age, heart rate, and BP;
6. validate externally.

48.4 Multi-site PPG

Simultaneous ear, finger, wrist, toe, or limb PPG can estimate intersite transit and detect asymmetry. This is more defensible than ECG-to-single-PPG for pure transit because it removes pre-ejection period, although path length, local vascular beds, and sensor delays remain.

Oscillometric Cuff Engineering

49.1 Pneumatic path

A research cuff system includes a pump, reservoir or cuff, fill valve, controlled deflation valve, emergency dump valve, pressure sensor, overpressure switch, and tubing. Safety should not depend on software alone. A mechanical or independent hardware pressure limit and timeout are necessary.

The pressure sensor needs adequate absolute accuracy and low noise across the full range, but oscillations may be only a small fraction of cuff pressure. The analog chain can use separate paths or digital processing to extract the slowly varying cuff pressure and pulsatile component.

49.2 Inflation and deflation

Inflate above estimated systolic pressure with a safe limit; deflate at a controlled rate or stepwise. Too-fast deflation undersamples beats and biases estimates. Too-slow deflation prolongs discomfort and venous congestion. Arrhythmia requires more beats and robust envelope construction.

49.3 Envelope algorithm development

A transparent research algorithm can:

1. remove slow cuff trend;
2. detect pulse oscillations and reject motion;
3. estimate amplitude versus cuff pressure;
4. fit a smooth envelope with uncertainty;
5. locate maximum amplitude for MAP proxy;
6. estimate SBP/DBP using ratios or learned features;
7. flag conditions outside training distribution.

Ratios should be calibrated on a development set and locked before validation. Clinical validation must be independent and follow ISO 81060-2 or the appropriate cuffless/continuous standard (85,95).

Mechanical and Acoustic Algorithm Design

50.1 Sensor fusion with ECG

ECG provides a stable electrical reference for segmentation. For each beat, extract synchronized windows from PCG, SCG, gyro, impedance, and PPG. Use robust templates and track quality. Do not force a fiducial when morphology lacks it.

50.2 Time-frequency analysis

Heart sounds and murmurs benefit from short-time Fourier, wavelet, or filter-bank analysis. Window length trades temporal and frequency resolution. Models should be tested against changes in microphone and chestpiece response. Augmentation with arbitrary noise does not substitute for real diverse recordings.

50.3 Mechanical transfer functions

An accelerometer measures enclosure acceleration. The measured waveform equals source motion convolved with tissue, attachment, enclosure, and sensor response. Repositioning by centimeters or changing adhesive can alter features. Validation should include repeated placement by multiple operators and a mechanical fixture for device-to-device comparison.

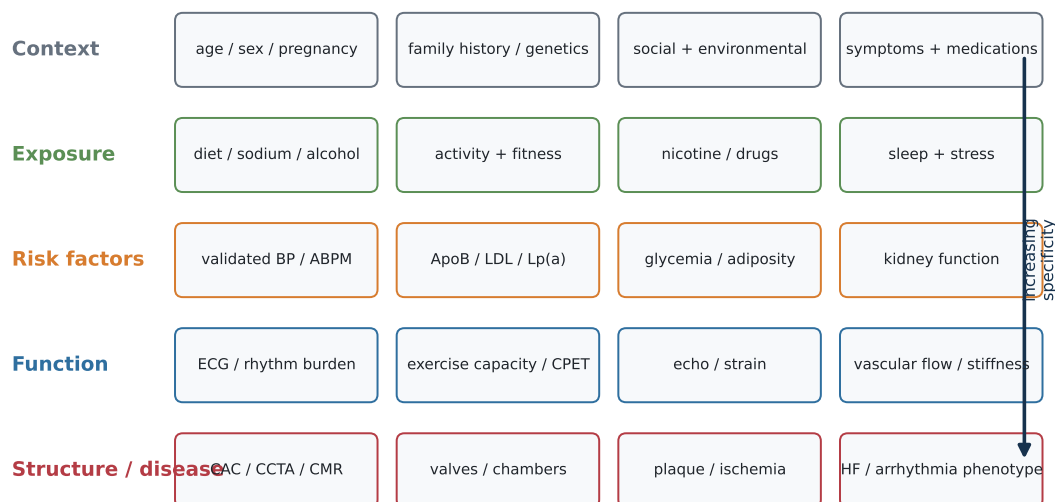
Multimodal Fusion and State Estimation

A multimodal cardiovascular monitor should not concatenate every feature into one opaque neural network as the first design. Begin with a physiological state model.

Possible latent states include:

- rhythm and beat validity;
- posture and activity;
- respiratory phase and rate;
- peripheral perfusion;
- vascular transit;
- electromechanical timing;
- sleep/wake state;
- congestion likelihood;
- deviation from personal baseline.

A complete cardiovascular profile is longitudinal and multimodal



Start broad and low-risk. Escalate when symptoms, baseline risk, or earlier measurements change the probability of disease.

No wearable can substitute for the clinical context and reference tests needed to answer a specific question.

Figure 51.1: Integrated profile: combine validated clinical measures, longitudinal wearable signals, context, and confirmatory escalation.

51.1 Hierarchical fusion

A robust hierarchy is:

1. per-channel calibration and quality;
2. beat or epoch detection;
3. within-modality features with uncertainty;
4. cross-modal timing and consistency checks;
5. context conditioning;

6. personal-baseline model;
7. population model;
8. alert logic and confirmatory recommendation.

A temperature rise, higher resting rate, lower activity, and altered respiration may indicate infection rather than cardiovascular decompensation. The system should represent competing explanations.

51.2 Personal baseline versus population threshold

Population thresholds are needed for clinical interpretation, but personal change can be more sensitive. A hybrid system uses both:

$$Score_t = f(Absolute_t, \Delta Personal_t, Context_t, Quality_t).$$

Baseline should adapt slowly and never absorb an unresolved disease transition. Model updates need guardrails, versioning, and rollback.

Data Architecture, Interoperability, and Security

52.1 Raw and derived layers

Use append-only raw files plus versioned derived tables.

Raw layer: exact ADC samples, device time, channel configuration, gain, LED current, quality/saturation flags, calibration, dropped packets.

Derived beat layer: fiducials, intervals, morphology features, confidence, rhythm labels.

Epoch layer: activity, posture, sleep state, respiratory rate, HRV, quality.

Session layer: protocol, device placement, symptoms, medications, reference-device identifiers.

Clinical layer: adjudicated labels, imaging/lab results, outcomes, provenance, access controls.

52.2 File formats

HDF5, Apache Parquet, EDF/EDF+, WFDB, DICOM, and FHIR each serve different roles. A practical architecture may store raw waveform chunks in HDF5 or WFDB, images in DICOM, analytics in Parquet, and clinical exchange through FHIR resources. Timestamps should be UTC with timezone and local context stored separately.

PhysioNet and PTB-XL demonstrate useful public ECG data structures and benchmarking resources (96,97). Public datasets are not sufficient for final product validation because device, population, care setting, and labeling differ.

52.3 Cybersecurity

Threat modeling should cover:

- unauthorized access to health data;
- firmware tampering;
- malicious commands to a therapeutic or measurement device;
- key extraction;
- insecure update channels;
- cloud account takeover;
- re-identification of waveform or location data;
- denial of service and battery drain.

Use secure boot, signed firmware, authenticated encryption, least privilege, key rotation, audit logs, vulnerability response, and an update policy. Cybersecurity is part of safety, not an IT afterthought (98).

Machine Learning: Valid Uses and Common Failure Modes

53.1 Appropriate roles

Machine learning is useful for:

- signal-quality assessment;
- beat and fiducial detection;
- denoising with preserved uncertainty;
- rhythm candidate detection;
- image acquisition guidance and segmentation;
- multimodal trend detection;
- prioritization for human review;
- individualized calibration within defined boundaries.

53.2 Dataset leakage

Leakage occurs when data from the same person appear in train and test sets, when post-outcome data are used, when device or hospital identity proxies labels, or when preprocessing uses the full dataset. Splits should be at patient and preferably site/time level.

53.3 Spectrum and prevalence bias

A model trained on obvious cases versus healthy controls performs worse in real clinics where disease is mild, mixed, treated, or confounded. Positive predictive value depends on prevalence:

$$PPV = \frac{Sensitivity \times Prevalence}{Sensitivity \times Prevalence + (1 - Specificity)(1 - Prevalence)}.$$

Even high specificity can generate many false positives in low-prevalence consumer screening.

53.4 Subgroup performance

Test age, sex, skin pigmentation, body size, rhythm, disease, medication, device fit, language, socioeconomic access, and site. Equal aggregate accuracy can hide subgroup failure. Calibration curves and false-negative consequences matter as much as area under the curve.

53.5 Model reporting and lifecycle

Follow TRIPOD for prediction models, STARD for diagnostic accuracy, CONSORT for trials, and PROBAST for risk-of-bias assessment (99–102). Medical-device ML should follow good machine-learning-practice principles, including representative data, independent test sets, human factors, monitoring, and controlled updates (103).

Safety, Human Factors, and Regulatory Framework

54.1 Risk management

ISO 14971 organizes hazard identification, risk estimation, controls, residual risk, and production feedback (104). Cardiovascular-monitor hazards include:

- shock or excessive leakage current;
- skin injury, pressure injury, allergy, or infection;
- cuff overpressure or prolonged occlusion;
- optical heating;
- battery thermal event;
- false negative causing delayed care;
- false positive causing anxiety or unnecessary treatment;
- wrong-patient data;
- timing or unit errors;
- algorithm bias;
- insecure update or data breach.

Risk control priority is inherently safe design, then protective measures, then information for safety. A warning label does not compensate for an avoidable electrical hazard.

54.2 Core standards map

Area	Representative standards / guidance
General electrical safety	IEC 60601-1 (105)
EMC	IEC 60601-1-2 (106)
Home environment	IEC 60601-1-11 (107)
Diagnostic ECG	IEC 60601-2-25 (76)
ECG monitoring	IEC 60601-2-27 (77)
Ambulatory ECG	IEC 60601-2-47 (78)
Invasive pressure	IEC 60601-2-34 (93)
Automated cuff BP	IEC 80601-2-30; ISO 81060-2 (84,85)
Continuous NIBP	ISO 81060-3 (95)
Pulse oximetry	ISO 80601-2-61 (89)
Software lifecycle	IEC 62304 (108)
Risk management	ISO 14971 (104)
Usability engineering	IEC 62366-1 (109)
Biological evaluation	ISO 10993 series (94)
Clinical investigation	ISO 14155 (110)
Rhythm algorithms	ANSI/AAMI EC57 (79)
Cybersecurity	FDA 2023 premarket guidance (98)

Standards evolve. The applicable edition, amendments, national deviations, classification, and predicate landscape must be confirmed at design freeze.

54.3 Human factors

Users misplace electrodes, wear straps too loosely, talk during BP, ignore charging, share devices, misunderstand notifications, and use devices outside labeled conditions. Usability engineering should test real tasks with representative users, including older adults, disability, low health literacy, color-vision differences, tremor, and stressful situations.

Avoid alarm language when the device only has a low-confidence estimate. Make “cannot analyze” distinct from “normal.” Preserve an emergency message for chest pain, severe breathlessness, syncope, or neurologic symptoms without claiming the device has excluded danger.

Verification and Validation Program

Validation is a ladder, not a single accuracy number

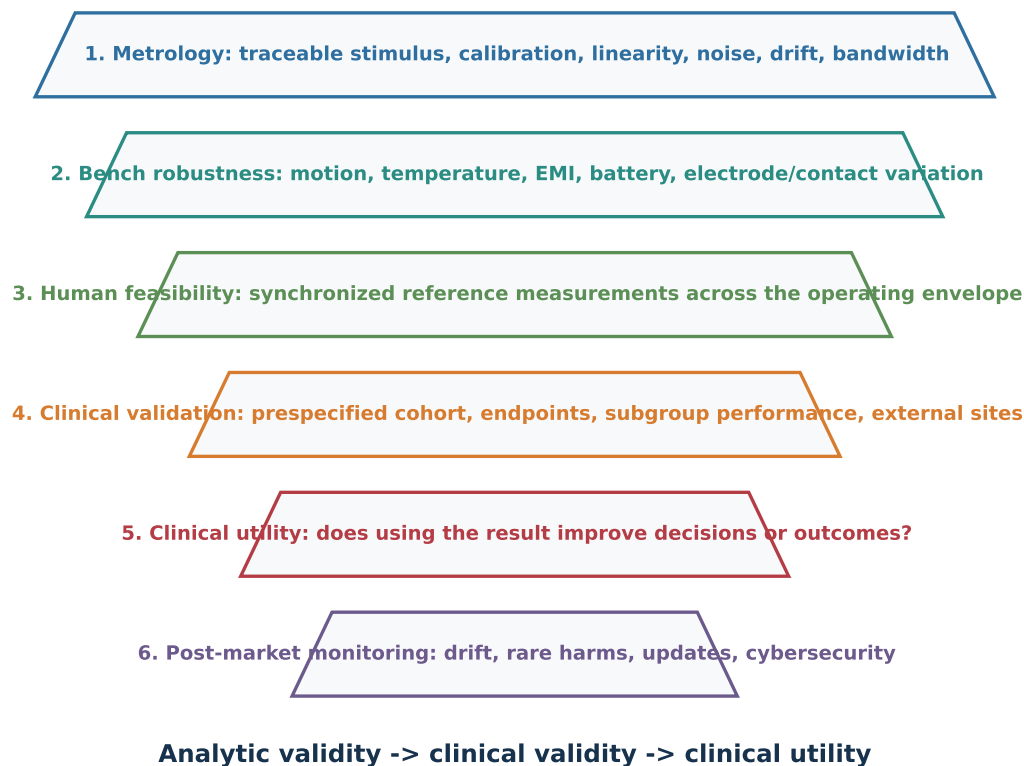


Figure 55.1: Validation ladder from component characterization through bench, human factors, analytical validation, clinical validation, and post-market monitoring.

55.1 Stage 0: requirements and hazards

Define intended use, users, environment, contraindications, outputs, reference methods, performance criteria, and unacceptable risks. Freeze a claim matrix before collecting confirmatory data.

55.2 Stage 1: component and subsystem characterization

- input-referred noise and bandwidth;
- ADC linearity and reference stability;
- electrode impedance tolerance;
- PPG optical power, crosstalk, ambient range, and thermal rise;
- pressure sensor accuracy, hysteresis, leak, and overpressure response;
- clock drift and synchronization;
- battery safety and current profile;
- mechanical and environmental tests.

55.3 Stage 2: simulator and benchtop verification

Use ECG/pacemaker/arrhythmia simulators, optical phantoms where appropriate, pressure calibrators, acoustic fixtures, motion rigs, temperature chambers, and programmable waveform sources. Simulators verify known waveforms and fault response but cannot replace human validation.

55.4 Stage 3: controlled healthy-volunteer feasibility

After safety review and consent, test posture, breathing, exercise, temperature, motion, sensor pressure, placement, and repeatability. The objective is to discover failure modes, not prove disease accuracy.

55.5 Stage 4: analytical validation

Compare device output with the appropriate reference under the full claimed range. Prespecify exclusions and quality handling. Analyze bias, limits of agreement, repeated measures, missingness, and subgroup performance. Lock the algorithm before the pivotal dataset.

55.6 Stage 5: clinical validation

Recruit the intended-use population, including difficult cases and realistic prevalence. Use independent adjudication. For diagnostic claims, report sensitivity, specificity, likelihood ratios, PPV/NPV, calibration, and decision consequences. For continuous measures, use agreement and clinical error zones where appropriate.

55.7 Stage 6: clinical utility

Show that using the device improves a meaningful process or outcome, not only detection. Does it reduce time to diagnosis, improve control, prevent admission, improve quality of life, or reduce unnecessary testing?

55.8 Stage 7: production and post-deployment monitoring

Verify device-to-device variation, manufacturing calibration, component changes, firmware updates, drift, complaints, cybersecurity, and real-world subgroup performance. A model update is a product change requiring controlled evaluation.

55.9 Statistical design essentials

- calculate sample size for the primary performance endpoint;
- account for repeated measures within participant;
- use an independent validation cohort;
- prespecify handling of missing and poor-quality data;
- report confidence intervals;
- avoid choosing thresholds after examining the test set;
- include all enrolled participants in a flow diagram;
- conduct sensitivity analyses for quality-gated versus all-attempt performance.

Reference Architecture A: ECG-PPG-IMU Chest Patch

56.1 Purpose

Research-quality synchronized acquisition of rhythm, pulse timing, chest mechanics, activity, respiration surrogates, and temperature over 24 hours to 7 days.

56.2 Suggested channels

- ECG: one differential channel plus reference/drive, 500 Hz or higher raw.
- PPG: red, infrared, and optionally green; 200-500 Hz with dark samples.
- IMU: 3-axis acceleration and gyro; 200-1000 Hz if SCG is a goal.
- Temperature: skin-contact plus internal board temperature.
- Optional: bioimpedance and contact microphone.

56.3 Outputs

- raw synchronized waveforms;
- beat locations and confidence;
- pulse rate, ECG rate, and disagreement flag;
- posture and activity;
- respiratory rate estimate with confidence;
- ECG-to-mechanical and ECG-to-PPG intervals;
- personal baseline trends;
- no unconfirmed diagnosis.

56.4 Validation

Use a reference ECG recorder, calibrated respiration, reference pulse oximeter when SpO₂ is claimed, and synchronized motion/video in controlled tests. Include walking, stairs, cycling, sleep, arm motion, ectopy, and varied skin/fit.

Reference Architecture B: Handheld or Wearable Six-Lead ECG

57.1 Purpose

Capture multiple limb vectors during symptoms or scheduled checks without full precordial setup.

57.2 Design

Use two or more contact electrodes and sequenced limb contact to derive I, II, III, aVR, aVL, and aVF. High input impedance and stable hand contact are essential. The device should display contact quality and record body configuration.

57.3 Limits

It does not provide V1-V6 and cannot replicate a standard 12-lead ECG. Chest pain evaluation, anterior/posterior ischemia, and many conduction or chamber patterns require standard leads and clinical assessment.

Reference Architecture C: Research Oscillometric Cuff

58.1 Purpose

Develop and validate transparent cuff-pressure algorithms while using a separate validated clinical device or dual-observer auscultation as reference.

58.2 Hardware

- interchangeable cuffs across intended arm range;
- pump, controlled valve, emergency dump;
- calibrated pressure sensor with traceability;
- independent overpressure cutoff;
- 200-500 Hz pressure acquisition;
- IMU or motion sensor on cuff;
- battery-powered operation and logged pneumatic state.

58.3 Development sequence

1. characterize pressure sensor and pneumatic leaks;
2. test with pressure calibrator and pulse simulator;
3. optimize deflation profile;
4. collect paired reference readings under a prespecified protocol;
5. lock envelope algorithm;
6. validate under ISO 81060-2-compatible conditions;
7. separately test arrhythmia, large arms, pregnancy, and other special populations before claiming them.

Reference Architecture D: Vascular Timing and ABI Station

59.1 Purpose

Measure upper- and lower-limb systolic pressures, Doppler waveforms, multi-site PPG, and pulse-wave timing under standardized posture.

59.2 Components

- validated cuffs with multiple sizes;
- continuous-wave Doppler with audio and waveform output;
- bilateral finger and toe PPG;
- ECG trigger;
- distance measurement protocol;
- temperature control and rest timer;
- software for ABI, toe-brachial index, pulse transit, asymmetry, and waveform quality.

59.3 Key risks

Sequential measurements drift with pressure and autonomic state. Simultaneous channels are preferable for timing. Calcified arteries can make ABI falsely high. PPG amplitude asymmetry can reflect temperature or contact, not obstruction. Diagnostic interpretation requires clinical protocol (48,49).

Reference Architecture E: Digital Stethoscope plus SCG

60.1 Purpose

Acquire synchronized ECG, heart sounds, and chest acceleration at standardized valve areas.

60.2 Hardware

- contact microphone or piezo element with calibrated response;
- tri-axial low-noise accelerometer/gyro;
- ECG trigger;
- controlled chestpiece pressure or force sensor;
- ambient microphone for noise reference;
- 4-8 kHz audio and 1-2 kHz mechanical acquisition.

60.3 Research outputs

- S1/S2 timing and confidence;
- systolic/diastolic spectral maps;
- electromechanical intervals;
- repeatability by site and operator;
- candidate murmur detection for research only.

Training data should include normal variants, lung sounds, obesity, prosthetic valves, arrhythmia, and real ambient noise.

Safe Laboratory Bring-Up

61.1 Equipment

- ECG and arrhythmia simulator;
- precision voltage/current source and DMM;
- oscilloscope used only on isolated/simulator setups unless medical isolation is verified;
- programmable power analyzer;
- pressure calibrator/manometer and pneumatic leak fixtures;
- optical test fixture and calibrated photodiode/power meter;
- temperature chamber or controlled hot/cold plate;
- motion shaker or repeatable mechanical fixture;
- audio calibrator and coupler;
- isolated trigger generator;
- ESD/EMC precompliance access as the design matures.

61.2 Bring-up checklist

1. Verify every rail with current limits before installing sensitive AFEs.
2. Measure quiescent and peak currents by subsystem.
3. Confirm references, clocks, ADC noise, and digital integrity with shorted or known inputs.
4. Verify lead protection and bias paths using a simulator.
5. Test saturation and recovery.
6. Characterize PPG dark current, ambient rejection, LED pulses, optical crosstalk, and thermal rise.
7. Verify clock drift over hours and days.
8. Exercise memory power loss, corruption recovery, and checksums.
9. Verify radio and flash activity do not corrupt analog channels.
10. Perform fault injection: open electrode, short, reversed sensor, blocked tube, leak, stuck valve, low battery, watchdog reset.
11. Complete schematic/ERC, PCB DRC, visual review, datasheet cross-check, and manufacturing package review before any production-readiness claim.

Budget every rail, provide test points and isolation links, separate analog and noisy current paths, preserve raw data, and make failure visible. A cardiovascular prototype that produces a plausible waveform while silently saturating, losing timestamps, or grounding the subject through a programmer is not a successful prototype.

Part VII

Integrated Cardiovascular Profiling and Research Roadmap

A Tiered Cardiovascular Health Profile

A complete profile should be proportional to the question and risk. Collecting every available signal can increase incidental findings, cost, and false positives without improving decisions.

62.1 Tier 0: Population and household context

Purpose: understand exposures that shape long-term risk before disease appears.

- age, sex, family and pregnancy history;
- tobacco/nicotine and alcohol;
- food access and dietary pattern;
- activity and sedentary time;
- sleep duration, regularity, and apnea symptoms;
- occupation, psychosocial stress, housing, finances, and care access;
- local air quality, heat exposure, and neighborhood environment.

Output: a contextual risk map, not a diagnostic score.

62.2 Tier 1: Minimal validated baseline

Purpose: establish major causal exposures and silent disease indicators.

- repeated validated upper-arm blood pressure;
- standard lipid panel, non-HDL-C, Lp(a), and apoB when useful;
- glucose/HbA1c;
- creatinine/eGFR and urine albumin-creatinine ratio;
- smoking status;
- weight, waist, and activity/fitness estimate;
- resting 12-lead ECG;
- symptom and family-history review;
- PREVENT short- and long-term risk where applicable.

This tier often provides more prevention value than a complex wearable. It measures the major treatable pathways directly.

62.3 Tier 2: Longitudinal home and wearable monitoring

Purpose: characterize dynamics and personal baseline.

- home BP protocol;
- synchronized ECG/PPG/IMU/temperature patch or watch;
- activity, resting heart rate, rhythm candidates, sleep timing, respiratory rate, and quality;
- symptom-event recordings;
- weight and optional validated SpO2;
- medication timing and selected context.

Output: trends, quality, and confirmatory prompts. No single composite “heart score” should obscure components.

62.4 Tier 3: Functional and structural assessment

Purpose: answer symptoms, abnormal examination, elevated risk, or research questions.

- echocardiography;
- ambulatory ECG or patch;
- ABPM;
- ABI/toe pressure and vascular ultrasound;
- exercise ECG, stress imaging, or CPET;
- coronary calcium or CCTA where appropriate;
- CMR, nuclear imaging, or targeted sleep testing.

62.5 Tier 4: Advanced invasive assessment

Purpose: diagnose and guide treatment in selected patients.

- coronary angiography with pressure/flow physiology;
- IVUS/OCT;
- right- and left-heart catheterization;
- electrophysiology study;
- biopsy;
- implantable pressure or rhythm monitoring.

This tier requires clinical indications, trained teams, sterile procedures, and regulated equipment.

A Standardized Longitudinal Monitoring Protocol

The protocol below is intended for research and device-development planning. It is not a substitute for clinician-directed testing.

63.1 Baseline week

1. **Enrollment and metadata:** demographics, medical history, family history, medications, device fit, informed consent, privacy choices.
2. **Clinical reference session:** standardized BP, 12-lead ECG, weight/waist, laboratory panel, symptom inventory.
3. **Device calibration:** electrode/site photos or diagrams, cuff size, clock synchronization, signal-quality confirmation.
4. **Controlled maneuvers:** five minutes supine, five minutes seated, standing transition, paced breathing, short walk, stairs or submaximal cycle if safe.
5. **Daily living:** at least three representative weekdays and one weekend day.
6. **Home BP:** morning and evening duplicate readings for a defined number of days with first-day handling specified.
7. **Symptom events:** user-initiated marker and a short description without interrupting raw recording.

63.2 Standard daily annotations

- wake/sleep time;
- medications and dose time;
- caffeine, alcohol, and major meals;
- structured exercise;
- unusual stress, illness, fever, or pain;
- symptoms;
- device removal, charging, poor fit, showering.

Annotations should be minimal enough to sustain adherence. Passive sensing should not be used to infer sensitive behavior without consent.

63.3 Weekly summaries

A credible summary contains:

- data coverage and quality by channel;
- resting heart-rate distribution under standardized states;
- rhythm candidates with raw strips;
- home BP average and variability with valid-reading count;
- activity and fitness trend;
- sleep duration/regularity and possible apnea flags;
- respiratory rate and temperature deviations;
- electromechanical timing research metrics with confidence;
- symptom correlations;
- unresolved device failures.

63.4 Escalation logic

Use a three-level approach:

- **Level A - device quality:** reposition, replace electrode, repeat cuff measurement, or withhold output.
- **Level B - confirmatory measurement:** obtain 12-lead ECG, validated BP, clinical pulse oximetry, laboratory test, or imaging.
- **Level C - urgent care:** symptoms or confirmed dangerous findings trigger established medical pathways; the device should not attempt autonomous treatment.

Data Schema for a Cardiovascular Research Platform

A minimal relational or event-based schema should include:

64.1 Participant table

- pseudonymous participant ID;
- consent version and permitted uses;
- demographics relevant to validation;
- medical and family history;
- pregnancy and reproductive history where relevant;
- device accessibility needs.

64.2 Device table

- serial number;
- hardware revision;
- sensor lot and calibration;
- firmware and algorithm version;
- battery age;
- maintenance and fault history.

64.3 Session table

- start/end UTC and local timezone;
- protocol version;
- sensor site and orientation;
- operator;
- reference devices;
- posture/activity schedule;
- environment;
- adverse events or deviations.

64.4 Waveform table or file index

- channel name and physical unit;
- nominal and measured sample rate;
- start timestamp and synchronization uncertainty;
- calibration and gain;
- missing samples and quality flags;
- file checksum and encryption status.

64.5 Annotation table

- event type;

- onset/offset;
- source: participant, device, technician, clinician, algorithm;
- confidence;
- adjudication status;
- linked raw-data window.

64.6 Outcome table

- reference diagnosis or measurement;
- method and device;
- adjudicators and blinding;
- date/time relation to index signal;
- uncertainty and missingness.

This structure enables reproducible reanalysis when algorithms change.

Practical Device Suite for a Development Laboratory

A staged suite avoids trying to build everything at once.

65.1 Suite 1: Reference acquisition

- research-grade multi-lead ECG recorder;
- validated clinical upper-arm BP monitor and cuffs;
- clinical pulse oximeter with waveform output if available;
- calibrated temperature and motion references;
- digital stethoscope or calibrated contact microphone;
- ECG/arrhythmia simulator and pressure calibrator.

65.2 Suite 2: Wearable development

- modular ECG/PPG/IMU/temperature board;
- battery logger with local storage;
- interchangeable electrode and optical mechanics;
- optical and motion fixtures;
- raw-data visualization and quality software.

65.3 Suite 3: Vascular station

- Doppler ultrasound probe;
- multiple cuffs and pressure channels;
- finger/toe PPG arrays;
- ECG trigger;
- standardized path-length and posture tools.

65.4 Suite 4: Functional testing

- instrumented cycle or treadmill;
- metabolic cart or validated CPET access;
- synchronized ECG/BP/SpO₂;
- exertion and symptom scales;
- emergency procedures and clinical supervision appropriate to risk.

65.5 Suite 5: External clinical partnerships

Rather than building CT, CMR, echo, catheterization, or laboratory assay systems, establish data-sharing and synchronization protocols with clinical facilities. The development value lies in paired reference data, not duplicating capital equipment.

Prioritized Build Roadmap

66.1 Phase 1: Safe waveform acquisition

Gate: clean, timestamped ECG, PPG, and IMU from simulators and controlled volunteers; complete rail/noise/clock characterization; no unisolated human connection.

Deliverables:

- hardware and firmware design files;
- raw-data format;
- calibration and test scripts;
- quality dashboards;
- documented failure modes.

66.2 Phase 2: Controlled physiology

Gate: repeatable signals across rest, posture, breathing, walking, exercise, and recovery against references.

Focus:

- sensor mechanics;
- timing synchronization;
- motion and temperature effects;
- within-person repeatability;
- quality gating.

66.3 Phase 3: Narrow analytical claims

Choose one or two claims, such as heart-rate accuracy during walking or ECG-to-PPG timing repeatability. Lock the algorithm and conduct an independent validation.

66.4 Phase 4: Disease cohorts and clinical partnership

Recruit representative arrhythmia, hypertension, heart-failure, PAD, or sleep cohorts under ethics approval. Pair signals with clinical reference tests and adjudication.

66.5 Phase 5: Utility study

Test whether the system changes a useful outcome: diagnostic yield, BP control, adherence, time to review, or reduction in unnecessary alerts.

The best first system is not a universal cardiovascular diagnostic. It is a transparent research platform that acquires synchronized raw ECG, PPG, motion, temperature, and optional acoustic/bioimpedance data; computes well-defined quality and timing measures; and integrates a validated external cuff. This platform creates the evidence needed to decide which narrower clinical product is defensible.

Open Questions and Research Frontiers

67.1 Calibration-free blood pressure

The field still lacks a universally validated, calibration-free, free-living cuffless solution across population, posture, temperature, exercise, vasoactive state, and disease. Progress may require direct local pressure sensing, ultrasound, multi-site transit, individualized biomechanical models, and periodic reference calibration rather than a single PPG feature.

67.2 Continuous cardiac output

Noninvasive cardiac-output estimation remains context-sensitive. Electrical cardiometry, bioimpedance, pulse contour, Doppler, and ballistocardiography can trend selected populations, but geometry, vascular tone, valve disease, and motion limit interchangeability. A useful research target is change detection under controlled interventions, not universal absolute output.

67.3 Multimodal foundation models

Large models may learn shared representations across ECG, PPG, acoustics, imaging, labs, and text. The risk is shortcut learning and opaque transfer across devices. Progress requires paired raw data, standardized timing, external validation, causal analysis, and model cards that describe failure.

67.4 Digital twins

Patient-specific cardiovascular models can integrate anatomy, pressure, flow, tissue, and control. Today, many parameters are unidentifiable from noninvasive data and model outputs depend on priors. Digital twins are most credible when used for constrained questions with sensitivity analysis rather than as complete virtual patients.

67.5 Wearable ultrasound and flexible sensors

Flexible arrays, stretchable electrodes, e-textiles, and skin-conformal optical systems can improve comfort and spatial sampling. Challenges are calibration under deformation, encapsulation, sweat, motion, sterilization, power, beam control, and manufacturing consistency.

67.6 Continuous biomarkers

Beyond glucose, continuous chemistry is limited by biofluid access, fouling, calibration, lag, and biological compartment differences. Cardiovascular applications may include electrolytes, lactate, drugs, inflammation, or coagulation, but clinical value must exceed the burden and risk of invasive access.

67.7 Preventive trials in younger adults

Lifetime risk models support earlier exposure reduction, but long outcome trials in young adults are difficult. Surrogates, genetics, subclinical imaging, and pragmatic health-system trials will remain important. The central unanswered question is not whether cumulative exposure matters, but how to deliver safe, equitable, durable prevention decades before events.

Final Synthesis

Cardiovascular health is best understood as a **dynamic reserve system under cumulative exposure**. The heart can maintain normal pressure and resting output despite substantial early disease. Symptoms often appear only when reserve is challenged or an acute trigger acts on a vulnerable substrate. That is why prevention relies heavily on measuring causal exposures - apoB particles, pressure, tobacco, glycemia, kidney disease, adiposity, activity, and sleep - before a wearable can see obvious pump failure.

Genetics matters substantially but does not erase actionability. Environment and social systems shape the exposure distribution. Nutrition, activity, sleep, stress, and substances act through measurable pathways. Medical treatment modifies causal biology. No percentage pie chart can represent these interactions faithfully.

The engineering corollary is equally important: a complete profile is not one sensor. It is a hierarchy of validated measures, longitudinal raw signals, quality control, context, and confirmatory escalation. Electrical, optical, mechanical, pressure, imaging, and biochemical modalities answer different questions. The most useful device is the one whose claim is narrow, whose failure is visible, whose reference is appropriate, and whose data lead to an effective action.

Core Cardiovascular Equations

A.1 Flow, resistance, and pressure

$$\Delta P = QR$$

$$R_{Poiseuille} = \frac{8\mu L}{\pi r^4}$$

$$MAP - RAP \approx CO \times SVR$$

For pulmonary circulation,

$$PVR = \frac{mPAP - PCWP}{CO}$$

with pressures commonly expressed in mmHg and cardiac output in L/min, producing Wood units.

A.2 Cardiac output and volume

$$CO = HR \times SV$$

$$SV = EDV - ESV$$

$$EF = \frac{EDV - ESV}{EDV}$$

$$CI = \frac{CO}{BSA}$$

A.3 Oxygen transport

$$C_{aO_2} = 1.34 \times Hb \times S_{aO_2} + 0.003 \times P_{aO_2}$$

$$DO_2 = CO \times C_{aO_2}$$

$$\dot{V}O_2 = CO(C_{aO_2} - C_{vO_2})$$

A.4 Wall stress

A simplified Laplace relation for a thin-walled cylindrical structure is

$$\sigma \approx \frac{Pr}{h},$$

showing why pressure and chamber radius increase wall stress while wall thickness reduces it. Ventricular geometry is not a thin cylinder, so this is intuition rather than exact calculation.

A.5 Compliance and elastance

$$C = \frac{\Delta V}{\Delta P}$$

$$E = \frac{\Delta P}{\Delta V}$$

Time-varying elastance models ventricular pressure as a function of volume and time, but real ventricles exhibit nonlinear, load- and history-dependent behavior.

A.6 Wave speed

$$PWV \approx \sqrt{\frac{Eh}{\rho D}}$$

or, using distensibility,

$$PWV = \sqrt{\frac{1}{\rho D_s}},$$

where D_s is arterial distensibility under the selected definition.

A.7 Doppler and pressure gradient

$$\Delta f = \frac{2f_0 v \cos \theta}{c}$$

$$\Delta P \approx 4v^2$$

A.8 Hydrostatic pressure

$$\Delta P = \rho gh$$

For blood, approximately 0.77 mmHg per centimeter vertical displacement.

A.9 Timing

$$PAT = PEP + PTT$$

$$PWV = \frac{\text{Path length}}{\text{Transit time}}$$

The path length must be anatomically defined and the timing fiducial consistent.

Reference Ranges and Interpretation Cautions

These are educational orientation ranges, not diagnostic rules. Laboratories, guidelines, age, sex, pregnancy, medications, posture, and clinical context alter interpretation.

Variable	Common orientation	Important caveat
Resting heart rate	Often 60-100 bpm in adults	Trained people may be lower; symptoms and rhythm matter.
BP category	Normal often below 120/80 mmHg; treatment decisions use broader context	Measurement technique and home/ambulatory patterns are essential (11).
LDL-C	Lower is generally better for atherosclerotic risk; goals depend on risk	ApoB, non-HDL-C, Lp(a), age, and disease alter targets (12).
Triglycerides	Fasting less than 150 mg/dL often considered normal	Nonfasting values and pancreatitis thresholds differ.
HbA1c	Less than 5.7% often normal screening range	Red-cell factors and assay limitations; diagnosis requires criteria.
eGFR	60 mL/min/1.73 m ² or higher often preserved	Age, albuminuria, trajectory, muscle mass, and non-steady state matter.
UACR	Less than 30 mg/g often normal/mild range	Confirm persistence; exercise/infection can elevate.
SpO ₂	Often 95%-100% at sea level	Altitude, disease, perfusion, device bias, and dyshemoglobin matter.
EF	Roughly 50%-70% often considered preserved	Method, loading, sex, and lab range; preserved EF does not exclude HF.
QTc	Often approximately less than 450 ms men / 460 ms women	Formula, rate, drugs, and lab criteria vary.
ABI	Approximately 1.00-1.40 commonly normal	0.91-0.99 borderline; above 1.40 may be noncompressible (48).

Modality Comparison Matrix

Modality	Physical measurand	Primary outputs	Invasiveness / setting	Strengths	Major failure modes	Best reference / validation
12-lead ECG	Body-surface potential differences	Rhythm, conduction, repolarization, morphology	Noninvasive clinic	Standardized views, rapid, inexpensive	Electrode misplacement, artifact, limited structural information	IEC performance plus expert interpretation
Ambulatory ECG patch	One/few voltage vectors over time	Rhythm burden, symptom correlation	Noninvasive ambulatory	Long duration, event capture	Skin/contact failure, limited vectors, false alerts	Paired reference ECG and adjudication
Consumer single-lead ECG	Contact voltage difference	User-triggered rhythm strip	Noninvasive consumer	Accessible, immediate documentation	Poor contact, limited lead, self-selection	Expert-reviewed ECG
PPG	Optical pulsatile tissue signal	Pulse rate, morphology, perfusion trends	Noninvasive wearable	Low cost, compact	Motion, contact, temperature, skin/optical bias	ECG for rate; calibrated references for other claims
Pulse oximetry	Dual/multi-wavelength pulsatile absorption	Estimated arterial oxygen saturation	Noninvasive	Continuous, mature	Low perfusion, motion, pigmentation bias, dyshemoglobin	Arterial co-oximetry under controlled protocol
Oscillometric cuff	Cuff pressure oscillations	SBP, MAP estimate, DBP	Noninvasive intermittent	Validated and scalable	Cuff size, motion, arrhythmia, proprietary ratios	ISO 81060-2 clinical validation
Finger volume clamp	Servo cuff pressure maintaining finger volume	Continuous finger pressure waveform	Noninvasive continuous	Beat-to-beat trends	Vasoconstriction, height, calibration, motion	Invasive pressure or appropriate reference
Tonometry	Surface pressure over flattened artery	Peripheral waveform, central estimate	Noninvasive research/clinic	Waveform morphology	Position/force, cuff calibration, transfer model	Invasive pressure in validation cohorts
PCG	Chest-surface sound/pressure	Heart sounds, murmurs, timing	Noninvasive	Low cost, familiar	Placement, ambient/lung noise, body habitus	Expert auscultation plus echo/hemodynamics
SCG/GCG	Chest acceleration/angular motion	Mechanical timing and morphology	Noninvasive wearable	Synchronized mechanics	Placement and attachment transfer function	Echo/PCG/pressure timing as appropriate
Bioimpedance	Complex voltage/current relationship	Fluid trends, respiration, cardiac waveform estimates	Noninvasive or implanted	Low power, continuous	Geometry, edema, motion, model dependence	Thermodilution/Fick or imaging for claimed output
Doppler ultrasound	Frequency shift of reflected ultrasound	Velocity and flow waveform	Noninvasive	Real-time physiology	Angle, sample placement, operator	Flow phantom and clinical reference
Echocardiography	Reflected ultrasound/image and Doppler	Structure, EF, valves, pressures, strain	Noninvasive clinic	Portable, broad information	Window, operator, loading, geometric assumptions	Core-lab protocol; CMR for selected volumes
Coronary calcium CT	X-ray attenuation of calcification	Calcified plaque score	Noninvasive imaging with radiation	Strong risk stratification	No noncalcified plaque or direct stenosis physiology	Standard acquisition/reconstruction and outcomes
CCTA	Contrast-enhanced X-ray tomography	Coronary lumen and plaque	Noninvasive imaging	High anatomic detail	Motion, calcification, contrast/radiation	CAD-RADS, invasive angiography where needed

Modality	Physical measurand	Primary outputs	Invasiveness / setting	Strengths	Major failure modes	Best reference / validation
CMR	MR signal, motion, flow, tissue relaxation	Volumes, flow, scar, edema, tissue	Noninvasive imaging	No ionizing radiation, tissue characterization	Access, duration, devices, protocol dependence	Standardized CMR protocol/core lab
SPECT/PET	Radiotracer emission	Perfusion, flow, metabolism, molecular targets	Noninvasive imaging with radiation	Functional/molecular information	Attenuation, relative perfusion, tracer logistics	Standard acquisition, outcomes, cross-modality reference
Arterial line	Fluid/solid-state transducer pressure	Continuous arterial waveform	Invasive critical care	Direct beat-to-beat pressure	Level/zero, damping, bubbles, catheter whip	Static calibration and dynamic response test
PA catheter	Intracardiac/pulmonary pressure, thermodilution	Filling pressures, CO, oxygen saturations	Invasive	Comprehensive hemodynamics	Waveform misread, wedge artifact, procedural risk	Expert protocol and Fick/thermodilution checks
FFR/iFR	Coronary pressure ratio	Lesion physiological significance	Invasive cath lab	Decision-linked physiology	Wire drift, damping, hyperemia/model assumptions	Trial-validated protocols
IVUS/OCT	Intravascular ultrasound/optical backscatter	Wall, plaque, stent dimensions	Invasive cath lab	High-resolution vessel information	Invasive risk, artifact, limited field/penetration	Standard calibration and expert core lab
Biomarker assay	Chemical concentration/activity	Lipids, injury, stress, metabolism	Blood/urine; invasive sampling	Mechanism-specific information	Preanalytics, assay specificity, biological variation	Traceable laboratory method and clinical cut-offs

Suggested Signal-Quality Metadata

Every waveform segment should carry machine-readable quality information.

Field	Example values
channel_status	valid, disconnected, saturated, clipped, calibration, unknown
contact_quality	impedance estimate, optical contact score, cuff fit
motion_level	RMS acceleration, posture, movement class
ambient_state	light level, temperature, electromagnetic interference flag
timing_quality	synchronized, drift-corrected, uncertain, gap
algorithm_quality	numeric confidence plus reason codes
exclusion_reason	motion, low perfusion, arrhythmia, missing reference, out-of-range
processing_version	firmware, filter, detector, model, calibration version

Quality should accompany every output rather than appearing only in a hidden debug log.

Human-Contact Prototype Safety Checklist

- Battery powered during contact.
- No unisolated USB, charger, oscilloscope, bench supply, or programmer while attached.
- Patient-lead current limiting and ESD/RF protection installed at connector.
- Bias and right-leg-drive currents limited under normal and single-fault conditions.
- Cuff has independent pressure limit, timeout, and dump path.
- Optical emitters assessed for thermal rise and duty-cycle fault.
- Enclosure prevents access to conductive internal parts.
- Skin materials documented and evaluated for contact type/duration.
- Data loss, saturation, lead-off, and poor quality are visible.
- Emergency language distinguishes measurement failure from normal result.
- Simulator tests completed before controlled human feasibility.
- Ethics, consent, privacy, and clinical oversight established for human research.
- Intended use explicitly excludes diagnosis/treatment until validated and authorized.

Glossary

- Afterload:** The load opposing ventricular ejection, including pressure, arterial impedance, geometry, and wall stress.
- ApoB:** Apolipoprotein B; one molecule is present on each major atherogenic lipoprotein particle, making plasma apoB a particle-number proxy.
- Arrhythmia burden:** Fraction of monitored time or number/duration of episodes in a rhythm, defined by a validated method.
- Calibration:** Mapping device output to a reference scale under stated conditions.
- Cardiac output:** Volume pumped per minute, equal to heart rate times stroke volume.
- Chronotropic competence:** Ability to increase heart rate appropriately with demand.
- Common-mode rejection:** Ability of a differential amplifier to reject voltage common to both inputs.
- Contractility:** Intrinsic active force-generating state of myocardium, conceptually independent of loading.
- Cuffless BP:** Blood-pressure estimate made without an occluding upper-arm cuff; often model- and calibration-dependent.
- Ejection fraction:** Stroke volume divided by end-diastolic volume.
- Electromechanical delay:** Time between electrical activation and mechanical event.
- Endothelial function:** Multiple vascular regulatory functions; no single test captures all of them.
- FFR:** Fractional flow reserve, a pressure-derived invasive index of coronary lesion significance under hyperemia.
- HFpEF:** Heart failure with preserved ejection fraction; a syndrome of elevated filling pressures and functional limitation despite preserved EF.
- HFrfEF:** Heart failure with reduced ejection fraction.
- HRV:** Heart-rate variability derived from ECG RR intervals; PPG pulse-rate variability is related but not identical.
- iFR:** Instantaneous wave-free ratio, a resting diastolic coronary pressure ratio.
- Lead:** An ECG voltage view; not synonymous with a physical electrode.
- Lp(a):** Lipoprotein(a), a genetically determined apoB-containing particle associated with atherosclerosis and aortic stenosis.
- MAP:** Mean arterial pressure, the time-average arterial pressure over a cardiac cycle.
- Measurand:** The physical or chemical quantity intended to be measured.
- Natriuretic peptides:** Hormones released with myocardial wall stress; BNP and NT-proBNP are clinically measured.
- PAT:** Pulse-arrival time from an electrical or other proximal reference to a peripheral pulse; often includes pre-ejection period.
- PEP:** Pre-ejection period from ventricular electrical activation to aortic valve opening.
- PPG:** Photoplethysmography, an optical measure of pulsatile tissue blood-volume-related change.
- Preload:** Initial filling/stretch condition before contraction; no single clinical measure captures it perfectly.
- PTT:** Pulse-transit time between two arterial waveform sites.
- PWV:** Pulse-wave velocity, path length divided by transit time; a measure related to arterial stiffness.
- Sensitivity:** Fraction of true positive cases detected.
- Specificity:** Fraction of true negative cases correctly rejected.
- Stroke volume:** End-diastolic volume minus end-systolic volume.

Troponin: Cardiac structural protein measured as a marker of myocardial injury; elevation is not synonymous with type 1 MI.

Validation: Demonstration that the device meets its intended-use performance in the intended population and conditions.

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